

Chapter Contents

2.1 Introduction

**2.2 Establishment of System Mathematical Model
(Differential Equation)**

**2.3 Solving Differential Equations Using the
Classical Time-Domain Method**

**2.4 Jump at the Starting Point - Conversion from
 0_- to 0_+ State**

2.5 Zero-Input Response and Zero-State Response

2.6 Impulse Response and Step Response

2.7 Convolution

2.8 Properties of Convolution

Contents of This Lecture

2.1 Introduction

2.2 Establishment of System Mathematical Model (Differential Equation)

2.3 Solving Differential Equations by the Classical Time-Domain Method

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

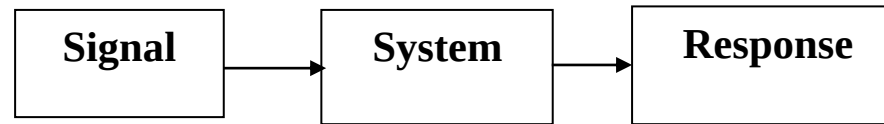
Learning Objectives of This Lecture

1. Familiarize with the method of establishing system differential equations (how to eliminate intermediate variables);
2. Proficiently master the classical method to solve the **free response** and **forced response** of the system;
3. Proficiently use the **impulse function matching method** to determine the jump variable of the starting point state.

Why Time-Domain Analysis of Systems?

Time-domain analysis of LTI systems offers:

- Intuitive physical insights, serving as the foundation for transform-domain analysis methods;
- Complex manual calculations, but software/tools (e.g., MATLAB) simplify solving.



Core Questions Addressed

For a given signal, a defined system, and the goal of finding the system response:

- How to establish the mathematical model of the system? (**Differential equations**)
- How to decompose the system response? (**Free response + Forced response; Zero-input response + Zero-state response**)
- How to solve different types of responses? (**Classical method & convolution method**)
- How to apply signals and systems? (**Simulation diagrams & computer-aided solving**)

2.2 Establishment of System Mathematical Model (Differential Equation)

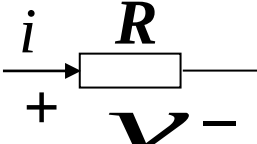


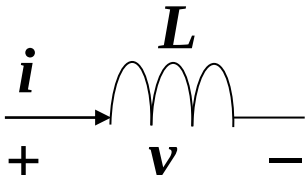
The goal of establishing a differential equation is to mathematically describe the physical characteristics of a system.

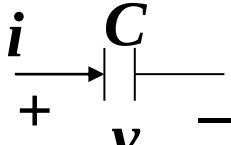
Electrical Circuit System Model

1) Component Constraints (Circuit Elements)

Voltage-Current Relationship

Resistor R  $v = Ri$

Inductor L  $i = \frac{1}{L} \int v dt$ $v = L di / dt$

Capacitor C  $i = C \frac{dv}{dt}$ $v = \frac{1}{C} \int i dt$

2) Network Topology Constraints

Kirchhoff's Current Law (KCL): At any instant, the sum of currents entering a node equals the sum of currents leaving the node:

$$\sum I_{in} = \sum I_{out}$$

Kirchhoff's Voltage Law (KVL): Around any closed loop, the sum of electromotive forces equals the sum of voltage drops across resistors:

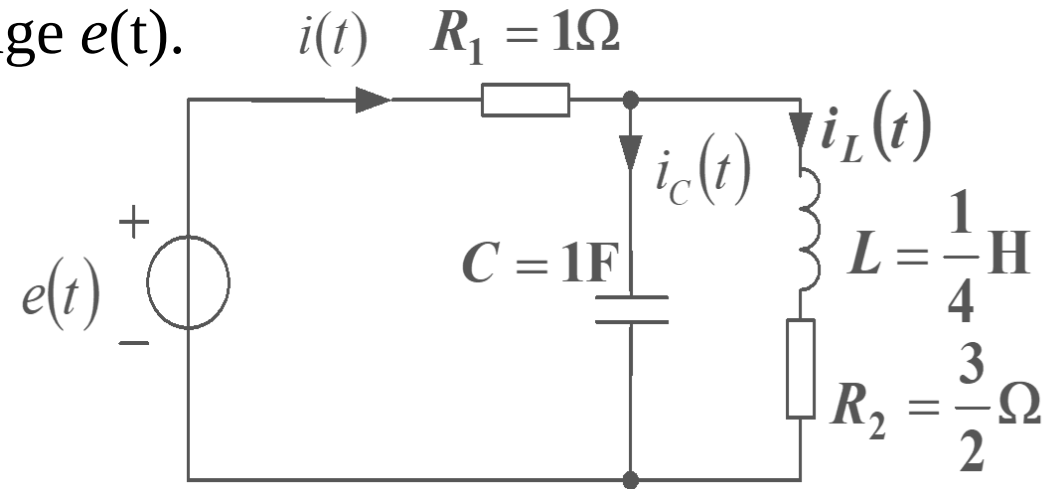
$$\sum_k E_k = \sum_k U_k$$

2.2 Establishment of System Mathematical Model (Differential Equation)



Establishing a Differential Equation for Circuit Current

Example 2-1: For the given circuit (as shown in the figure), establish the differential equation for current $i(t)$ with input voltage $e(t)$.



Solution:

List KVL/KCL equations:

$$\text{Loop voltage equation: } R_1 i(t) + v_C(t) = e(t) \quad (1)$$

$$\text{Loop voltage equation: } v_C(t) = L \frac{d}{dt} i_L(t) + i_L(t) R_2 \quad (2)$$

$$\text{Node current equation: } i(t) = C \frac{d}{dt} v_C(t) + i_L(t) \quad (3)$$

2.2 Establishment of System Mathematical Model (Differential Equation)



Eliminate intermediate variables, i.e., $v_C(t)$ and $i_L(t)$.

$$R_1 i(t) + v_C(t) = e(t) \quad (1)$$

(1) is simplest, containing $v_C(t)$ only, not $v_C'(t)$.

$$v_C(t) = L \frac{d}{dt} i_L(t) + i_L(t) R_2 \quad (2)$$

(2) is most complex, containing $i_L'(t)$ and related to $v_C''(t)$.

$$i(t) = C \frac{d}{dt} v_C(t) + i_L(t) \quad (3)$$

(3) Contains $v_C'(t)$.

$$\text{From (1), derive } v_C(t) = e(t) - R_1 i(t) \quad (4)$$

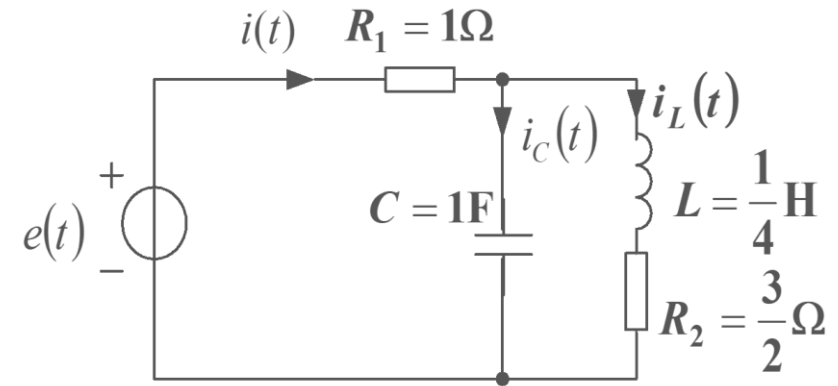
$$\text{Substituting (4) into (3), derive } i_L(t) = i(t) + C R_1 \frac{d}{dt} i(t) - C \frac{d}{dt} e(t) \quad (5)$$

Substitute (4) and (5) into (2), resulting in a single differential equation for $i(t)$ and $e(t)$ ((6) on next page).

2.2 Establishment of System Mathematical Model (Differential Equation)



$$\begin{aligned} & \frac{d^2}{dt^2} i(t) + \left(\frac{1}{R_1 C} + \frac{R_2}{L} \right) \frac{d}{dt} i(t) + \left(\frac{1}{LC} + \frac{R_2}{R_1 LC} \right) i(t) \\ &= \frac{1}{R_1} \frac{d^2}{dt^2} e(t) + \frac{R_2}{R_1 L} \frac{d}{dt} e(t) + \frac{1}{R_1 LC} e(t) \end{aligned} \quad (6)$$



Substituting the component parameters into (6) yields

$$\frac{d^2}{dt^2} i(t) + 7 \frac{d}{dt} i(t) + 10 i(t) = \frac{d^2}{dt^2} e(t) + 6 \frac{d}{dt} e(t) + 4 e(t)$$

For complex systems, the relationship between the input $x(t)$ and output $y(t)$ is described by an **n -th order linear constant-coefficient ordinary differential equation (ODE)**:

$$\begin{aligned} a_n \frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) \\ = b_m \frac{d^m x(t)}{dt^m} + b_{m-1} \frac{d^{m-1} x(t)}{dt^{m-1}} + \dots + b_1 \frac{dx(t)}{dt} + b_0 x(t) \end{aligned}$$

The **complete solution** of this ODE consists of two parts:

- 1) **Homogeneous solution**: Also called **free response**, it describes the system's inherent behavior (determined by the system itself).
- 2) **Particular solution**: Also called **forced response**, it describes the system's behavior under the influence of the input (determined by the input signal).

2.3 Solving Differential Equations by the Classical Time-Domain Method

Step 1: Solve the Homogeneous Solution

The homogeneous solution satisfies the **homogeneous ODE** (set the input side to 0):

$$a_n \frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = 0$$

Characteristic Equation: $a_n \alpha^n + a_{n-1} \alpha^{n-1} + \dots + a_1 \alpha + a_0 = 0$

The roots $\alpha_1 \dots \alpha_n$ are called the system's “natural frequencies”.

Roots of the Characteristic Equation	Form of Homogeneous Solution (Coefficients $A_1, \dots, A_n$ are undetermined)
Distinct real roots	$y_h(t) = A_1 e^{\alpha_1 t} + A_2 e^{\alpha_2 t} + \dots + A_n e^{\alpha_n t}$
Repeated real roots	If α_1 is a k -th order repeated root, the corresponding term is $(A_1 t^{K-1} + A_2 t^{K-2} + \dots + A_{K-1} t + A_K) e^{\alpha_1 t}$
Complex conjugate roots	If $\alpha_{1,2} = \alpha \pm j\beta$, the corresponding term is $e^{\alpha t} (A_1 \cos \beta t + A_2 \sin \beta t)$

2.3 Solving Differential Equations by the Classical Time-Domain Method

Step 2: Solve the Particular Solution

The form of the particular solution depends on the input signal (or the “**free term**” of the ODE, i.e., the right-hand side of the ODE).

Free Term	Form of Particular Solution $y_p(t)$
E (constant)	B (constant)
t^p	$B_p t^p + B_{p-1} t^{p-1} + \dots + B_1 t + B_0$
$e^{\alpha t}$	$\begin{cases} Be^{\alpha t} & (\text{If } \alpha \text{ is not a characteristic root.}) \\ Bt^k e^{\alpha t} & (k: \text{multiplicity of } \alpha) \end{cases}$
$\cos \omega_0 t / \sin \omega_0 t$	$B_1 \cos \omega_0 t + B_2 \sin \omega_0 t$

Step 3: Find Undetermined Coefficients

Substitute the complete solution $\mathbf{y}(t) = \mathbf{y}_h(t) + \mathbf{y}_p(t)$ into the original ODE, and solve for the undetermined coefficients A_1, A_2, A_n using the initial conditions (0_+ states).

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Definition of States

1. **0_- state**: The system state just before the input is applied (no input influence).
2. **0_+ state**: The system state just after the input is applied (used as initial conditions for solving the ODE, since **the ODE solution is valid for $t \geq 0_+$**).

Circuit Switching Rule

For circuit systems, the 0_- state describes the energy storage of components (capacitor voltage $v_C(0_-)$, inductor current $i_L(0_-)$).

In most cases, **the voltage across a capacitor and the current through an inductor do not change abruptly during switching**: $v_C(0_-) = v_C(0_+)$, $i_L(0_-) = i_L(0_+)$

However, **if an impulse current is applied to a capacitor or an impulse voltage to an inductor, $i_C(t)$ or $v_L(t)$ will jump between 0_- and 0_+** .

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

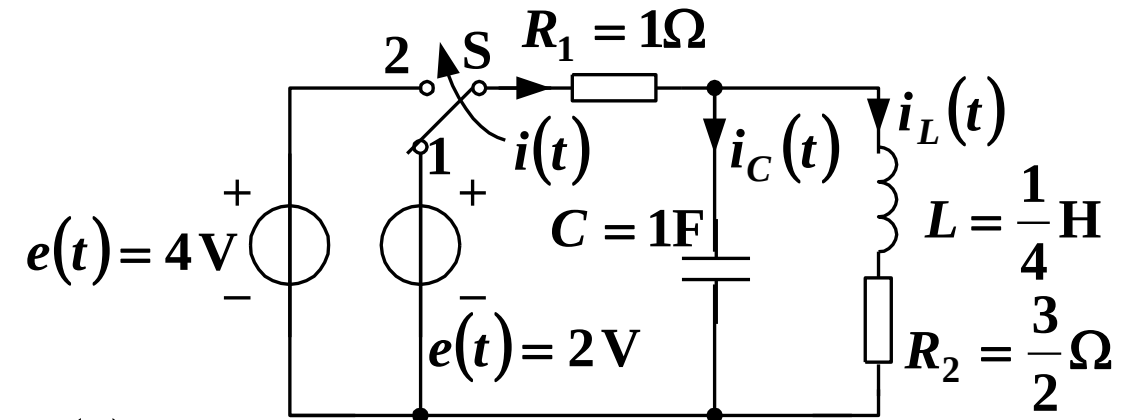
Example 2-2: Solve the differential equation established in Example 2-1. Assume that the switch moves from 1 to 2 when $t=0$.

Solution:

The differential equation established is

$$\frac{d^2}{dt^2} i(t) + 7 \frac{d}{dt} i(t) + 10i(t) = \frac{d^2}{dt^2} e(t) + 6 \frac{d}{dt} e(t) + 4e(t)$$

The input signal can be expressed as $e(t) = 2u(t) + 2 \text{ V}$.



Step 1: Find the homogeneous solution

- 1) Characteristic equation: $\alpha^2 + 7\alpha + 10 = 0 \rightarrow$ Roots $\alpha_1 = -2, \alpha_2 = -5$;
- 2) Homogeneous solution: $i_h(t) = A_1 e^{-2t} + A_2 e^{-5t}$, where A_1 and A_2 are undetermined.

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Step 2: Find the particular solution

- 1) Substitute the input $e(t)=2u(t)+2$ in the right side of the differential equation and **keep only the nonzero terms when $t>0$,**

$$\frac{d^2}{dt^2} i(t) + 7 \frac{d}{dt} i(t) + 10i(t) = \frac{d^2}{dt^2} e(t) + 6 \frac{d}{dt} e(t) + 4e(t)$$



$$\frac{d^2}{dt^2} i(t) + 7 \frac{d}{dt} i(t) + 10i(t) = 16$$

- 2) Assume $i_p(t) = B$;
- 3) Substitute $i_p(t)=B$ into the ODE, $10B=16 \rightarrow B = 1.6$.

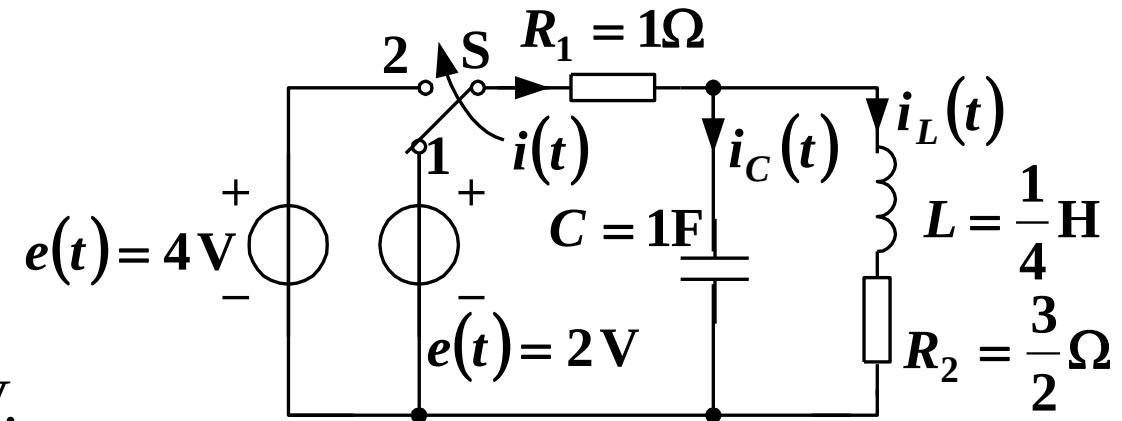
2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Step 3: Complete solution

$$i(t) = i_h(t) + i_p(t) = A_1 e^{-2t} + A_2 e^{-5t} + 1.6$$

When $t=0_-$, $i(0_-) = e(0_-) / (R_1 + R_2) = 0.8$ A,

$$i_L(0_-) = i(0_-) = 0.8 \text{ A}, \quad V_C(0_-) = i_L(0_-) R_2 = 1.2 \text{ V}.$$



As the voltage across a capacitor and the current through an inductor do not change from $t=0_-$ to 0_+ , $V_C(0_+) = V_C(0_-) = 1.2$ V, $i_L(0_+) = i_L(0_-) = 0.8$ A.

$$i(0_+) = \frac{1}{R_1} [e(0_+) - v_C(0_+)] = 4 - 1.2 = 2.8 \text{ A}$$

$$\begin{aligned} \frac{d}{dt} i(0_+) &= \frac{1}{R_1} \left[\frac{d}{dt} e(0_+) - \frac{d}{dt} v_C(0_+) \right] = \frac{1}{R_1} \left[0 - \frac{i_C(0_+)}{C} \right] \\ &= \frac{i_L(0_+) - i(0_+)}{R_1 C} = i_L(0_-) - i(0_+) = 0.8 - 2.8 = -2 \text{ A/s} \end{aligned}$$

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Step 4: Determine Coefficients A_1 and A_2

$$As \quad i(t) = A_1 e^{-2t} + A_2 e^{-5t} + 1.6 \quad \frac{d}{dt} i(t) = -2A_1 e^{-2t} - 5A_2 e^{-5t}$$

$$\begin{cases} i(0_+) = A_1 + A_2 + 1.6 = 2.8 \\ \frac{d}{dt} i(0_+) = -2A_1 - 5A_2 = -2 \end{cases} \quad \rightarrow \quad \begin{cases} A_1 = \frac{4}{3} \\ A_2 = -\frac{2}{15} \end{cases}$$

The complete solution is

$$i(t) = \left(\frac{4}{3} e^{-2t} - \frac{2}{15} e^{-5t} + \frac{8}{5} \right) A \quad (t > 0)$$

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Example 2-3: For the given circuit (as shown in the figure), input $x(t)=\cos(2t)u(t)$, initial voltages of both capacitors are zero. Find the response $v_2(t)$.

Solution:

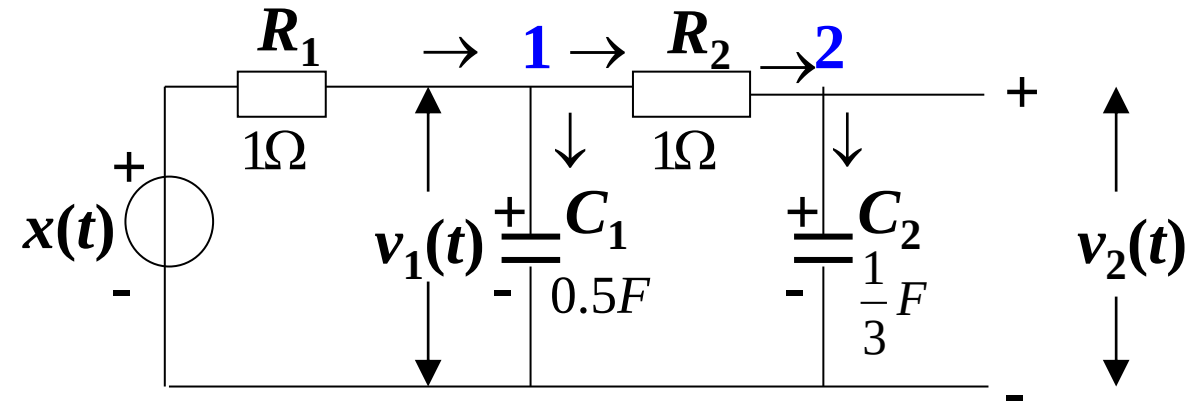
Step 1: Establish the ODE

List KCL equations for nodes 1 and 2,

$$\begin{cases} \frac{x(t) - v_1(t)}{R_1} = C_1 \frac{dv_1(t)}{dt} + \frac{v_1(t) - v_2(t)}{R_2} & (1) \\ \frac{v_1(t) - v_2(t)}{R_2} = C_2 \frac{dv_2(t)}{dt} & (2) \end{cases}$$

From (2), derive $v_1(t) = R_2 C_2 \frac{dv_2(t)}{dt} + v_2(t)$ (3)

Substituting (3) into (1), derive the ODE $\frac{d^2 v_2(t)}{dt^2} + 7 \frac{dv_2(t)}{dt} + 6v_2(t) = 6x(t)$



2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

$$\text{ODE: } \frac{d^2 v_2(t)}{dt^2} + 7 \frac{dv_2(t)}{dt} + 6v_2(t) = 6x(t)$$

Step 2: Find the homogeneous solution

$$\text{Characteristic equation: } \alpha^2 + 7\alpha + 6 = 0 \quad \text{Roots: } \alpha_1 = -1, \alpha_2 = -6$$

$$\text{Homogeneous solution: } A_1 e^{-t} + A_2 e^{-6t} \quad (A_1 \text{ and } A_2 \text{ are undetermined})$$

Step 3: Find the particular solution

Substitute $x(t) = \cos(2t)$ ($t > 0$) into the right side of the ODE,

$$\frac{d^2 v_2(t)}{dt^2} + 7 \frac{dv_2(t)}{dt} + 6v_2(t) = 6 \cos 2t$$

so assume the particular solution is $B_1 \sin 2t + B_2 \cos 2t$. Substitute it into the left of the ODE,

$$(2B_1 - 14B_2) \sin 2t + (14B_1 + 2B_2) \cos 2t = 6 \cos 2t$$

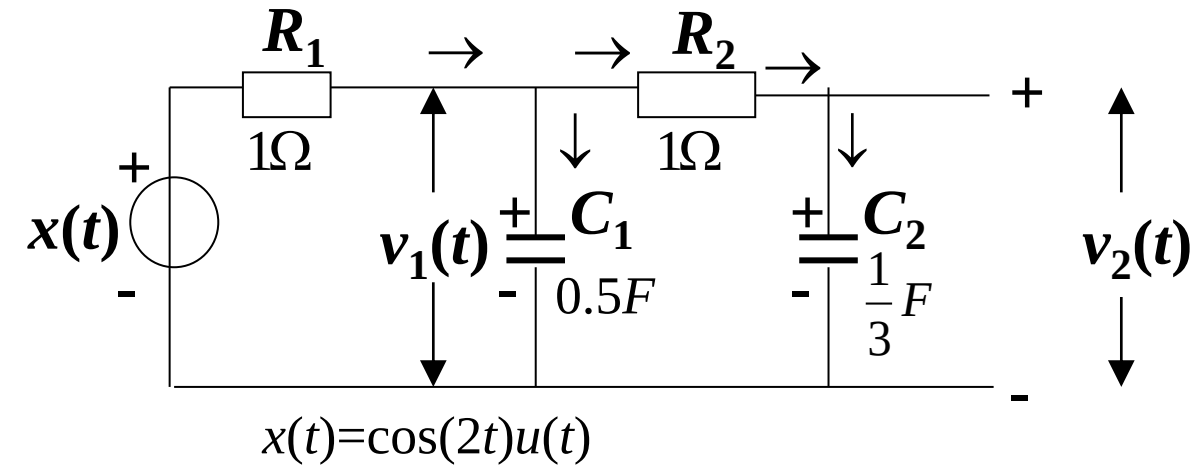
$$\begin{cases} 2B_1 - 14B_2 = 0 \\ 14B_1 + 2B_2 = 6 \end{cases} \longrightarrow B_1 = \frac{21}{50}, B_2 = \frac{3}{50}$$

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Step 4: Find the complete solution

$$v_2(t) = A_1 e^{-t} + A_2 e^{-6t} + \frac{21}{50} \sin 2t + \frac{3}{50} \cos 2t \quad (t > 0)$$

$$\frac{dv_2(t)}{dt} = -A_1 e^{-t} - 6A_2 e^{-6t} + \frac{21}{25} \cos 2t - \frac{3}{25} \sin 2t$$



As $v_2(0_-) = 0$, and the voltage across C_2 does not change abruptly at $t=0$, hence $v_2(0_+) = 0$; Similarly, $v_1(0_-) = v_1(0_+) = 0$. The voltage across R_2 remains at 0 from 0_- to 0_+ , and so there is no current through R_2 and C_2 . Hence, $\frac{dv_2(0_+)}{dt} = 0$.

With $v_2(0_+) = 0$, $A_1 + A_2 = -\frac{3}{50}$

With $\frac{dv_2(0_+)}{dt} = 0$, $A_1 + 6A_2 = \frac{21}{25}$



$$A_1 = -\frac{6}{25}, A_2 = \frac{9}{50}$$

Complete solution: $v_2(t) = -\frac{6}{25} e^{-t} + \frac{9}{50} e^{-6t} + \frac{21}{50} \sin(2t) + \frac{3}{50} \cos(2t) \quad (t > 0)$

- Repeat the examples in the lecture notes.
- 2.52(a) and 2.53(b) in the textbook.

Review of Last Lecture

2.1 Introduction

**2.2 Establishment of System Mathematical Model
(Differential Equation)**

**2.3 Solving Differential Equations by the Classical
Time-Domain Method**

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

2.2 Establishment of System Mathematical Model (Differential Equation)

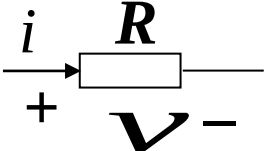


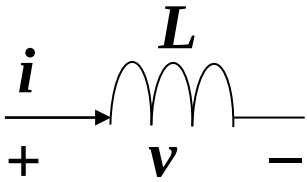
The goal of establishing a differential equation is to mathematically describe the physical characteristics of a system.

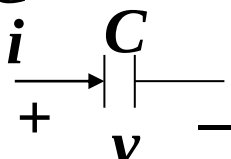
Electrical Circuit System Model

1) Component Constraints (Circuit Elements)

Voltage-Current Relationship

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Inductor L  $i = \frac{1}{L} \int v dt$ $v = L di / dt$

Capacitor C  $i = C \frac{dv}{dt}$ $v = \frac{1}{C} \int i dt$



2) Network Topology Constraints

Kirchhoff's Current Law (KCL): At any instant, the sum of currents entering a node equals the sum of currents leaving the node:

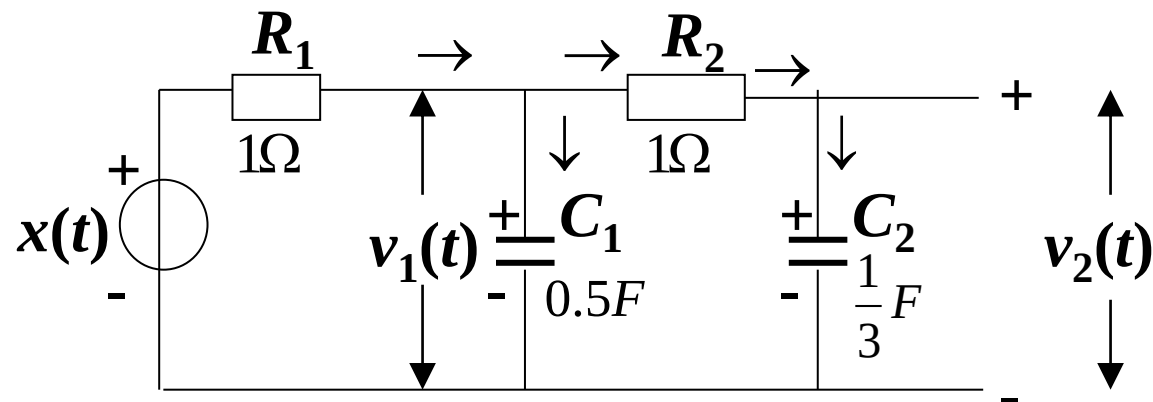
$$\sum I_{in} = \sum I_{out}$$

Kirchhoff's Voltage Law (KVL): Around any closed loop, the sum of electromotive forces equals the sum of voltage drops across resistors:

$$\sum_k E_k = \sum_k U_k$$

Assume that the complete response of the circuit system shown in the figure, with excitation $x(t)=\cos(2t)u(t)$, is $v_2(t)=-\frac{6}{25}e^{-t}+\frac{9}{50}e^{-6t}+\frac{21}{50}\sin(2t)+\frac{3}{50}\cos(2t)$ ($t > 0$). The forced response is ().

- A $-\frac{6}{25}e^{-t}+\frac{9}{50}e^{-6t}$ ($t > 0$)
- B $\frac{21}{50}\sin(2t)$ ($t > 0$)
- C $\frac{3}{50}\cos(2t)$ ($t > 0$)
- D $\frac{21}{50}\sin(2t)+\frac{3}{50}\cos(2t)$ ($t > 0$)**



For complex systems, the relationship between the input $x(t)$ and output $y(t)$ is described by an **n -th order linear constant-coefficient ordinary differential equation (ODE)**:

$$\begin{aligned} a_n \frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) \\ = b_m \frac{d^m x(t)}{dt^m} + b_{m-1} \frac{d^{m-1} x(t)}{dt^{m-1}} + \dots + b_1 \frac{dx(t)}{dt} + b_0 x(t) \end{aligned}$$

The **complete solution** of this ODE consists of two parts:

- 1) **Homogeneous solution**: Also called **free response** or **natural response**, it describes the system's inherent behavior (determined by the system itself).
- 2) **Particular solution**: Also called **forced response**, it describes the system's behavior under the influence of the input (determined by the input signal).

2.3 Solving Differential Equations by the Classical Time-Domain Method

Step 1: Solve the Homogeneous Solution

The homogeneous solution satisfies the **homogeneous ODE** (set the input side to 0):

$$a_n \frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = 0$$

Characteristic Equation: $a_n \alpha^n + a_{n-1} \alpha^{n-1} + \dots + a_1 \alpha + a_0 = 0$

The roots $\alpha_1 \dots \alpha_n$ are called the system's “natural frequencies”.

Roots of the Characteristic Equation	Form of Homogeneous Solution (Coefficients $A_1, \dots, A_n$ are undetermined)
Distinct real roots	$y_h(t) = A_1 e^{\alpha_1 t} + A_2 e^{\alpha_2 t} + \dots + A_n e^{\alpha_n t}$
Repeated real roots	If α_1 is a k -th order repeated root, the corresponding term is $(A_1 t^{K-1} + A_2 t^{K-2} + \dots + A_{K-1} t + A_K) e^{\alpha_1 t}$
Complex conjugate roots	If $\alpha_{1,2} = \alpha \pm j\beta$, the corresponding term is $e^{\alpha t} (A_1 \cos \beta t + A_2 \sin \beta t)$

2.3 Solving Differential Equations by the Classical Time-Domain Method

Step 2: Solve the Particular Solution

The form of the particular solution depends on the input signal (or the “**free term**” of the ODE, i.e., the right-hand side of the ODE).

Free Term	Form of Particular Solution $y_p(t)$
E (constant)	B (constant)
t^p	$B_p t^p + B_{p-1} t^{p-1} + \dots + B_1 t + B_0$
$e^{\alpha t}$	$\begin{cases} Be^{\alpha t} & (\text{If } \alpha \text{ is not a characteristic root.}) \\ Bt^k e^{\alpha t} & (k: \text{multiplicity of } \alpha) \end{cases}$
$\cos \omega_0 t / \sin \omega_0 t$	$B_1 \cos \omega_0 t + B_2 \sin \omega_0 t$

Step 3: Find Undetermined Coefficients

Substitute the complete solution $\mathbf{y}(t) = \mathbf{y}_h(t) + \mathbf{y}_p(t)$ into the original ODE, and solve for the undetermined coefficients A_1, A_2, A_n using the initial conditions (0_+ states).

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Definition of States

1. **0_- state**: The system state just before the input is applied (no input influence).
2. **0_+ state**: The system state just after the input is applied (used as initial conditions for solving the ODE, since **the ODE solution is valid for $t \geq 0_+$**).

Circuit Switching Rule

For circuit systems, the 0_- state describes the energy storage of components (capacitor voltage $v_C(0_-)$, inductor current $i_L(0_-)$).

In most cases, **the voltage across a capacitor and the current through an inductor do not change abruptly during switching**: $v_C(0_-) = v_C(0_+)$, $i_L(0_-) = i_L(0_+)$

However, **if an impulse current is applied to a capacitor or an impulse voltage to an inductor, $i_C(t)$ or $v_L(t)$ will jump between 0_- and 0_+** .

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Example 2-3: For the given circuit (as shown in the figure), input $x(t)=\cos(2t)u(t)$, initial voltages of both capacitors are zero. Find the response $v_2(t)$.

Solution:

Step 1: Establish the ODE

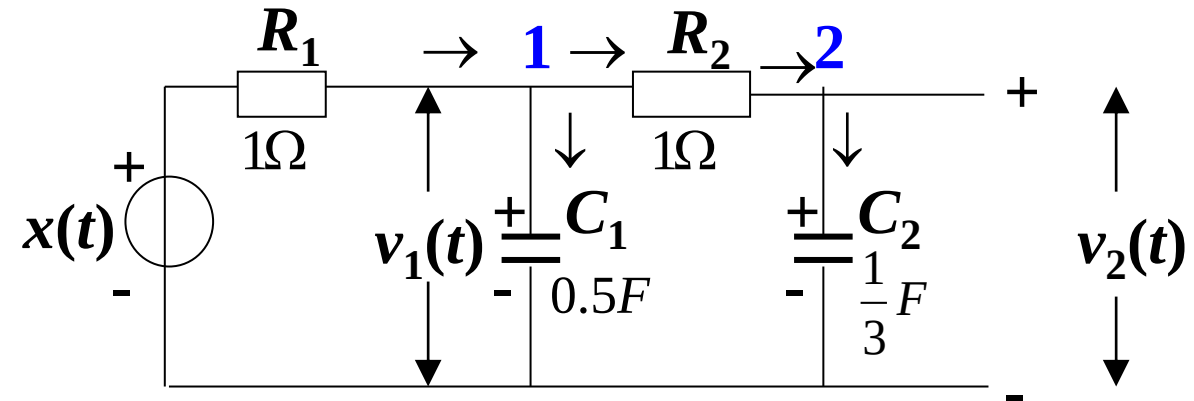
List KCL equations for nodes 1 and 2,

$$\left\{ \begin{array}{l} \frac{x(t) - v_1(t)}{R_1} = C_1 \frac{dv_1(t)}{dt} + \frac{v_1(t) - v_2(t)}{R_2} \end{array} \right. \quad (1)$$

$$\left\{ \begin{array}{l} \frac{v_1(t) - v_2(t)}{R_2} = C_2 \frac{dv_2(t)}{dt} \end{array} \right. \quad (2)$$

From (2), derive $v_1(t) = R_2 C_2 \frac{dv_2(t)}{dt} + v_2(t)$ (3)

Substituting (3) into (1), derive the ODE $\frac{d^2 v_2(t)}{dt^2} + 7 \frac{dv_2(t)}{dt} + 6v_2(t) = 6x(t)$



2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

$$\text{ODE: } \frac{d^2 v_2(t)}{dt^2} + 7 \frac{dv_2(t)}{dt} + 6v_2(t) = 6x(t)$$

Step 2: Find the homogeneous solution

$$\text{Characteristic equation: } \alpha^2 + 7\alpha + 6 = 0 \quad \text{Roots: } \alpha_1 = -1, \alpha_2 = -6$$

$$\text{Homogeneous solution: } A_1 e^{-t} + A_2 e^{-6t} \quad (A_1 \text{ and } A_2 \text{ are undetermined})$$

Step 3: Find the particular solution

Substitute $x(t) = \cos(2t)$ ($t > 0$) into the right side of the ODE,

$$\frac{d^2 v_2(t)}{dt^2} + 7 \frac{dv_2(t)}{dt} + 6v_2(t) = 6 \cos 2t$$

so assume the particular solution is $B_1 \sin 2t + B_2 \cos 2t$. Substitute it into the left of the ODE,

$$(2B_1 - 14B_2) \sin 2t + (14B_1 + 2B_2) \cos 2t = 6 \cos 2t$$

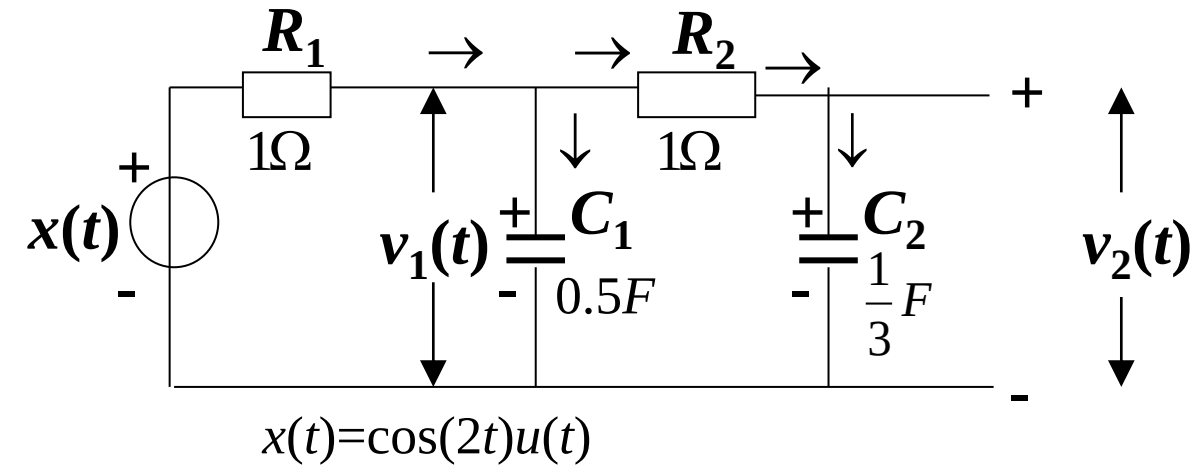
$$\begin{cases} 2B_1 - 14B_2 = 0 \\ 14B_1 + 2B_2 = 6 \end{cases} \longrightarrow B_1 = \frac{21}{50}, B_2 = \frac{3}{50}$$

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Step 4: Find the complete solution

$$v_2(t) = A_1 e^{-t} + A_2 e^{-6t} + \frac{21}{50} \sin 2t + \frac{3}{50} \cos 2t \quad (t > 0)$$

$$\frac{dv_2(t)}{dt} = -A_1 e^{-t} - 6A_2 e^{-6t} + \frac{21}{25} \cos 2t - \frac{3}{25} \sin 2t$$



As $v_2(0_-) = 0$, and the voltage across C_2 does not change abruptly at $t=0$, hence $v_2(0_+) = 0$; Similarly, $v_1(0_-) = v_1(0_+) = 0$. The voltage across R_2 remains at 0 from 0_- to 0_+ , and so there is no current through R_2 and C_2 . Hence, $\frac{dv_2(0_+)}{dt} = 0$.

With $v_2(0_+) = 0$, $A_1 + A_2 = -\frac{3}{50}$

With $\frac{dv_2(0_+)}{dt} = 0$, $A_1 + 6A_2 = \frac{21}{25}$



$$A_1 = -\frac{6}{25}, A_2 = \frac{9}{50}$$

Complete solution: $v_2(t) = -\frac{6}{25} e^{-t} + \frac{9}{50} e^{-6t} + \frac{21}{50} \sin(2t) + \frac{3}{50} \cos(2t) \quad (t > 0)$

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Decomposition of the Complete Response:

Free/Natural Response (Homogeneous Solution): System's inherent behavior, from homogeneous ODE (input = 0), determined by natural frequencies and initial energy.

Forced Response (Particular Solution): System's behavior driven by input, form matches input (e.g., sinusoidal for sinusoidal input).

$$v_2(t) = \underbrace{-\frac{6}{25}e^{-t} + \frac{9}{50}e^{-6t}}_{\text{Free response}} + \underbrace{\frac{21}{50}\sin(2t) + \frac{3}{50}\cos(2t)}_{\text{Forced response}}$$

Free response
(homogeneous solution) **Forced response**
(particular solution)

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Decomposition of the Complete Response:

When the input signal is a step signal or a causal periodic signal, the complete response of a stable system can be divided into **transient response** and **steady-state response**.

The **transient response** refers to the response that appears temporarily in the complete response after the excitation is applied, and gradually fades away as time increases.

The **steady-state response** usually consists of step functions and periodic functions.

$$v_2(t) = \underbrace{-\frac{6}{25}e^{-t} + \frac{9}{50}e^{-6t}}_{\text{Transient response}} + \underbrace{\frac{21}{50}\sin(2t) + \frac{3}{50}\cos(2t)}_{\text{Steady-state response}}$$

Transient response **Steady-state response**

Contents of This Lecture

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State (Continued)

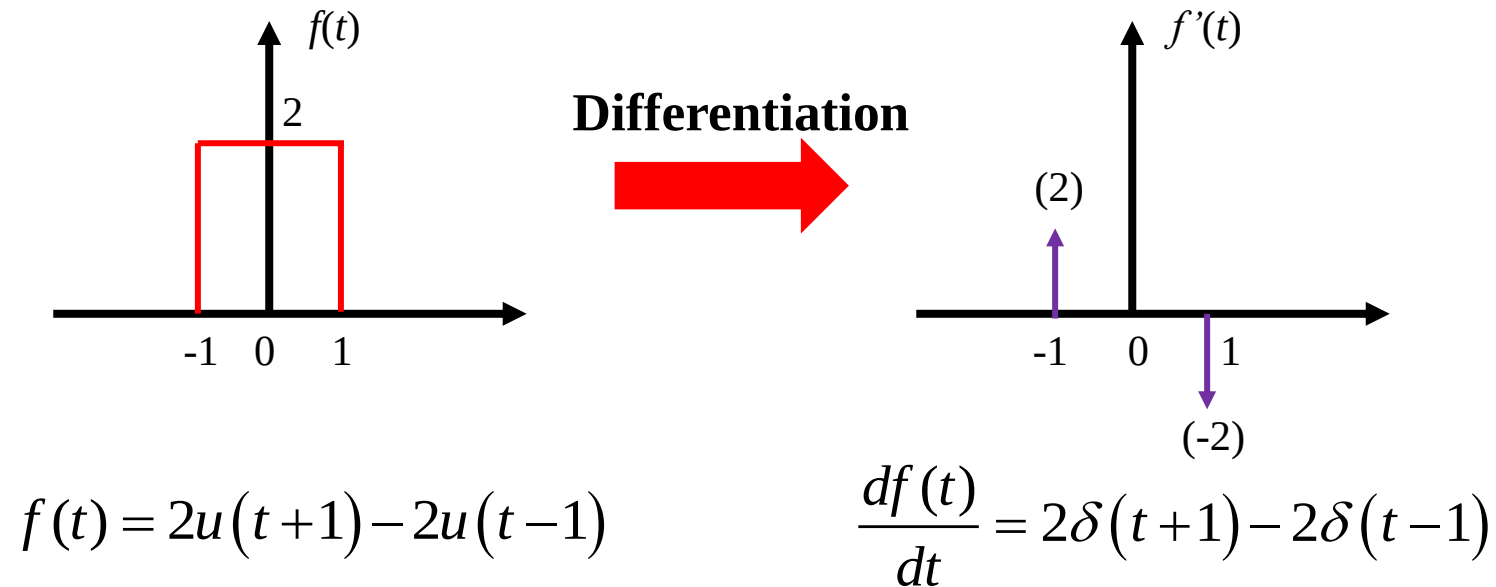
2.5 Zero-Input Response and Zero-State Response

Learning Objectives of This Lecture

1. Proficiently use the **impulse function matching method** to determine the jump variable of the starting point state;
2. Master the **classical method** to solve the **zero-input response** (homogeneous solution) of a system;
3. Proficiency in using the **classical method** and the **impulse function matching method** to solve the **zero-state response**;

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Impulse Function Matching Method: Impulse functions are the cause of state jumps (for example, an impulse current in a capacitor leads to a jump in its voltage). Therefore, **at $t=0$, the $\delta(t)$ term and its derivatives of all orders on both sides of the differential equation must be balanced.**



If there is a state jump, its derivative must be an impulse function.

In general, the coefficients of the homogeneous solution can be solved by the impulse function matching method.

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Example 2-4: Given $\frac{d}{dt}r(t) + 3r(t) = 3\delta'(t)$, and $r(0_-) = 1$, find $r(0_+)$.

Solution: Since the right-hand side of the equation has $3\delta'(t)$, it must be part of $\frac{d}{dt}r(t)$.

$$\text{Assume } \frac{d}{dt}r(t) = 3\delta'(t) + a\delta(t)$$

$$r(t) = 3\delta(t) + \boxed{a\Delta u(t)} \quad \leftarrow \text{It means } r(t) \text{ jumps by } a \text{ from } 0_- \text{ to } 0_+.$$

$\Delta u(t)$: **Relative Unit Jump Function** – State jump by 1 from 0_- to 0_+

Substitute into the equation, and balance impulse terms

$$3\delta'(t) + (a + 9)\delta(t) = 3\delta'(t) \quad \longrightarrow \quad a + 9 = 0 \quad \longrightarrow \quad a = -9$$

$$r(0_+) = r(0_-) + a = 1 - 9 = -8$$

Given ODE $r''(t) + 3r'(t) + 2r(t) = 2\delta(t)$, $r(0_-)$ and $r'(0_-)$,
Find $r(0_+)$ and $r'(0_+)$.

- ☐ A $r(0_+) = r(0_-)$, $r'(0_+) = r'(0_-)$
- ☒ B $r(0_+) = r(0_-)$, $r'(0_+) = r'(0_-) + 2$
- ☐ C $r(0_+) = r(0_-) + 2$, $r'(0_+) = r'(0_-)$
- ☐ D $r(0_+) = r(0_-) + 3$, $r'(0_+) = r'(0_-) + 2$

提交

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Example 2-5: Given $r''(t) + 3r'(t) + 2r(t) = 2\delta(t)$, $r(0_-)$ and $r'(0_-)$, find $r(0_+)$ and $r'(0_+)$.

Solution:

Method 1: Assume $r''(t) = a\delta(t)$, $r'(t) = a\Delta u(t)$, $r(t) = 0$ ($t=0$)

$$a\delta(t) + 3a\Delta u(t) = 2\delta(t) \longrightarrow a = 2$$

$$r'(0_+) = r'(0_-) + 2 \qquad r(0_+) = r(0_-)$$

Method 2: The term $2\delta(t)$ in the right-hand side must belongs to $r''(t)$, and it means that **there is a jump of +2 at $r'(t)$ from 0_- to 0_+** .

$$r'(0_+) = r'(0_-) + 2$$

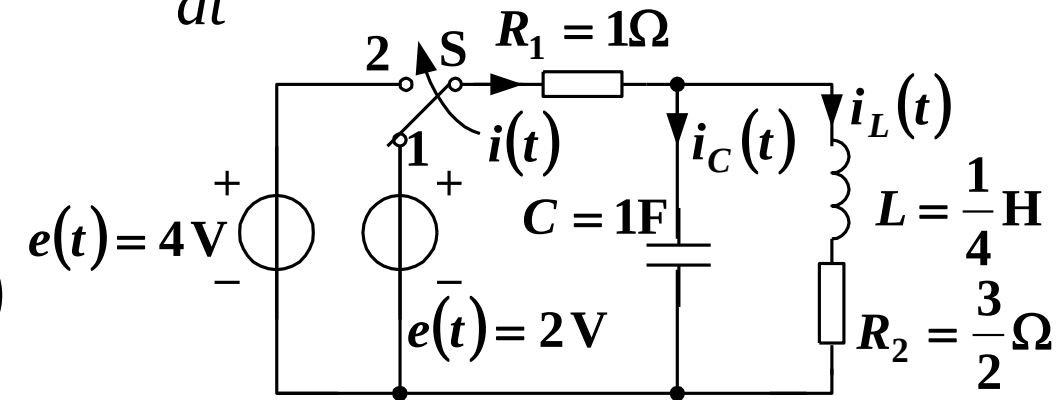
$$r(0_+) = r(0_-)$$

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Use the Impulse Function Matching Method to find the state changes from 0_- to 0_+ in Example 2-2. Given the circuit shown in the figure, switch S is in position 1 and has reached a steady state. When $t=0$, it switches from position 1 to position 2. Establish the differential equation for current $i(t)$ and determine $i(0_+)$ and $\frac{d}{dt}i(0_+)$ after the switching.

Solution: The differential equation of the circuit:

$$\frac{d^2}{dt^2}i(t) + 7\frac{d}{dt}i(t) + 10i(t) = \frac{d^2}{dt^2}e(t) + 6\frac{d}{dt}e(t) + 4e(t)$$



At around $t=0_+$, $e(t) = e(0_-) + 2\Delta u(t) = 2 + 2\Delta u(t)$,

Substitute it, $\frac{d^2}{dt^2}i(t) + 7\frac{d}{dt}i(t) + 10i(t) = 2\delta'(t) + 12\delta(t) + 8\Delta u(t) + 8$

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

$$\frac{d^2}{dt^2} i(t) + 7 \frac{d}{dt} i(t) + 10i(t) = 2\delta'(t) + 12\delta(t) + 8\Delta u(t) + 8$$

Assume $\frac{d^2}{dt^2} i(t) = 2\delta'(t) + a\delta(t)$, then $\frac{d}{dt} i(t) = 2\delta(t) + \underline{a}\Delta u(t)$, $i(t) = \underline{2}\Delta u(t)$

State jump amount from 0_- to 0_+

Balance the impulse functions and their derivatives,

$$2\delta'(t) + (a+14)\delta(t) = 2\delta'(t) + 12\delta(t)$$

$$a+14=12 \Rightarrow a=-2$$

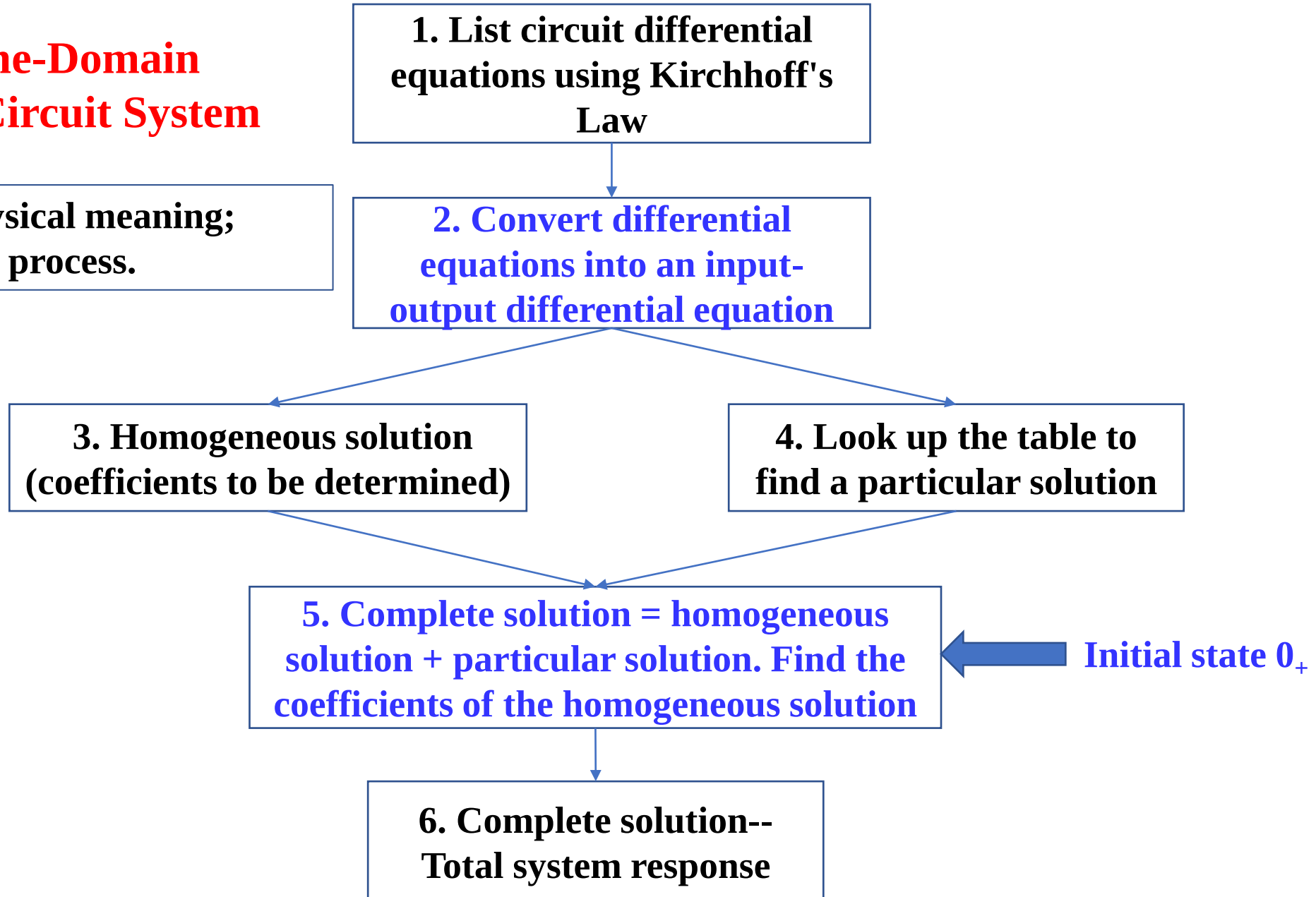
$$\therefore i(0_+) = i(0_-) + 2 = 0.8 + 2 = 2.8 \text{ A}, \quad \frac{d}{dt} i(0_+) = \frac{d}{dt} i(0_-) - 2 = 0 - 2 = -2 \text{ A/s}$$

The coefficients of the homogeneous solution can then be solved with the states at 0_+ .

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Classical Time-Domain Analysis of Circuit System

Pros: Clear physical meaning;
Cons: Complex process.



2.5 Zero-Input Response and Zero-State Response

- The complete response of a system solved by the classical method can be divided into:

Complete Response = Free Response + Forced Response

$$r(t) = r_h(t) + r_p(t)$$

- The complete response of the system can also be divided into:

Complete Response = Zero-Input Response + Zero-State Response

$$r(t) = r_{zi}(t) + r_{zs}(t)$$

2.5 Zero-Input Response and Zero-State Response

1. Definition of Zero-Input Response and Determination of Undetermined Coefficients

① Definition: The response generated entirely by the initial state without the action of an external excitation signal, i.e., $r_{zi}(t) = H[\{x(0_-)\}]$

② Equation Satisfied:
$$c_0 \frac{d^n}{dt^n} r_{zi}(t) + \dots + c_{n-1} \frac{d}{dt} r_{zi}(t) + c_n r_{zi}(t) = 0$$

Therefore, it is in the form of a homogeneous solution, i.e.,

$$r_{zi}(t) = \sum_{k=1}^n A_{zik} e^{\alpha_k t}$$

where $\alpha_1, \alpha_2, \dots, \alpha_n$ are n distinct system characteristic roots.

③ Initial Conditions: $r_{zi}^{(k)}(0_+) = r_{zi}^{(k)}(0_-) = r^{(k)}(0_-)$

That is, the undetermined coefficients of the homogeneous solution $r_{zi}(t)$ can be determined by $r^{(k)}(0_-)$.

2.5 Zero-Input Response and Zero-State Response

2. Definition of Zero-State Response and Determination of Undetermined Coefficients

- ① Definition: The response generated only by the excitation when the initial state is zero.

$$r_{zs}(t) = H[e(t)]$$

- ② Equation Satisfied:
$$c_0 \frac{d^n}{dt^n} r_{zs}(t) + \dots + c_{n-1} \frac{d}{dt} r_{zs}(t) + c_n r_{zs}(t) = E_0 \frac{d^m}{dt^m} c(t) + \dots + E_m$$

Therefore, $r_{zs}(t) = \sum_{k=1}^n A_{zsk} e^{\alpha_k t} + r_p(t)$, i.e., it contains both the homogeneous solution and the particular solution.

- ③ Initial Conditions: Since the k -th ($k = 0, \dots, n-1$) order derivatives of $r_{zs}(t)$ at time 0_- are all zero, i.e.,

$$r^{(k)}(0_-) = r_{zi}^{(k)}(0_-) + r_{zs}^{(k)}(0_-) = r_{zi}^{(k)}(0_-)$$

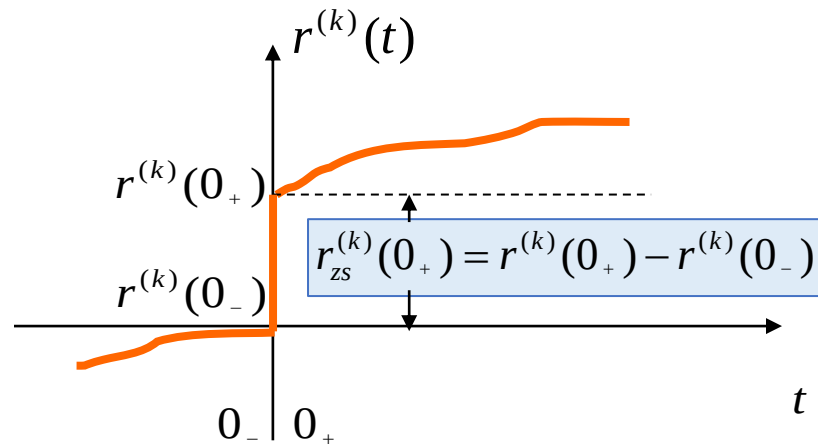
$$r_{zs}^{(k)}(0_-) = 0$$

$$r^{(k)}(0_+) = r_{zi}^{(k)}(0_+) + r_{zs}^{(k)}(0_+) = r_{zi}^{(k)}(0_-) + r_{zs}^{(k)}(0_+)$$

$$r_{zs}^{(k)}(0_+) = r^{(k)}(0_+) - r^{(k)}(0_-) = \text{Jump value}$$

Thus, A_{zsk} is determined by the jump value.

2.5 Zero-Input Response and Zero-State Response



$r^{(k)}(0_+)$: Used to determine the coefficients of the **complete response**

$r^{(k)}(0_-)$: Used to determine the coefficients of the **zero-input response**

$r_{zs}^{(k)}(0_+)$: Used to determine the coefficients of the **zero-state response**

$$\begin{aligned}
 r(t) &= \underbrace{\sum_{k=1}^n A_{Zik} e^{a_k t}}_{\text{zero-input response}} + \underbrace{\sum_{k=1}^n A_{Zsk} e^{a_k t}}_{\text{zero-state response}} + r_p(t) \\
 &= \underbrace{\sum_{k=1}^n (A_{Zik} + A_{Zsk}) e^{a_k t}}_{\text{free response}} + \underbrace{r_p(t)}_{\text{forced response}}
 \end{aligned}$$

Important!

Which of the following statements is **incorrect**? ()

- ☐ A The zero-state response of a system includes part of the natural response and the forced response.
- ☒ B If the initial state of the system is zero, the system response is the forced response of the system.
- ☐ C The zero-state response is independent of the initial state of the system and is generated by the excitation signal.
- ☐ D The zero-input response is independent of the system excitation and is generated by the initial state of the system.

提交

2.5 Zero-Input Response and Zero-State Response

Key Facts

- **Free Response:** Corresponding to the system characteristics;
- **Forced Response:** Corresponding to the excitation signal.
- **Zero-Input Response:** Caused by the internal energy storage of the system;
- **Zero-State Response:** Caused by the excitation signal.
- For many practical problems in communication and electronic systems, only the **zero-state response** needs to be calculated; to find the zero-state response, the cumbersome classical method is not required, and instead, the **convolution method** can be used.

2.5 Zero-Input Response and Zero-State Response

Example 2-6: Given the differential equation of a system $r''(t) + 3r'(t) + 2r(t) = 2e'(t) + 6e(t)$, the initial state $r(0_-) = 2$, $r'(0_-) = 0$, and $e(t) = u(t)$, find the zero-input response and zero-state response of the system.

Solution: (1) Find the zero-input response $r_{zi}(t)$

$$r_{zi}''(t) + 3r_{zi}'(t) + 2r_{zi}(t) = 0$$

Finding the initial value: $r_{zi}(0_+) = r_{zi}(0_-) = r(0_-) = 2$

$$r_{zi}'(0_+) = r_{zi}'(0_-) = r'(0_-) = 0$$

With the characteristic roots -1 and -2, assume $r_{zi}(t) = A_1 e^{-t} + A_2 e^{-2t}$

Substitute the initial values to find the coefficients $A_1 = 4$, $A_2 = -2$

$$r_{zi}(t) = 4e^{-t} - 2e^{-2t} \quad (t > 0)$$

2.5 Zero-Input Response and Zero-State Response

Example 2-6: Given the differential equation of a system $r''(t) + 3r'(t) + 2r(t) = 2e'(t) + 6e(t)$, the initial state $r(0_-) = 2$, $r'(0_-) = 0$, and $e(t) = u(t)$, find the zero-input response and zero-state response of the system.

(2) Find the zero-state response $r_{zs}(t)$

$$r_{zs}''(t) + 3r_{zs}'(t) + 2r_{zs}(t) = 2\delta(t) + 6u(t)$$

$$r_{zs}(0_+) = 0 \quad r_{zs}'(0_+) = r_{zs}'(0_-) + 2 = 2$$

It is observed that $r_{zs}(0)$ does not jump, while $r_{zs}'(0)$ jumps by 2.

$$\text{When } t > 0, \quad r_{zs}''(t) + 3r_{zs}'(t) + 2r_{zs}(t) = 6$$

Assume the homogeneous solution is: $r_{zsh}(t) = C_1 e^{-t} + C_2 e^{-2t}$

Assume the particular solution is a constant B. Substitute it into the equation:
 $2B = 6$, so $B = 3$.

Complete solution: $r_{zs}(t) = C_1 e^{-t} + C_2 e^{-2t} + 3$ Its derivative: $r_{zs}'(t) = -C_1 e^{-t} - 2C_2 e^{-2t}$

Substitute the initial states $r_{zs}(0_+) = 0$ and $r_{zs}'(0_+) = 2$, to find $C_1 = -4$ and $C_2 = 1$.

Therefore, $r_{zs}(t) = -4e^{-t} + e^{-2t} + 3 \quad (t > 0)$

- Repeat the examples in the lecture notes.
- 2.55(a), 2.57(a)(c) and 2.58(a)(c) in the textbook.
- Additional questions on **the Impulse Function Matching Method**:

$$(1) \quad 2 \frac{d^2 r(t)}{dt^2} + 3 \frac{dr(t)}{dt} + 4r(t) = \frac{de(t)}{dt}$$

Given $e(t) = u(t)$, $r(0_-) = 1$, $r'(0_-) = 1$, find $r(0_+)$ and $r'(0_+)$.

Solution: $r(0_+) = 1, r'(0_+) = \frac{3}{2}$

$$(2) \quad \frac{d^2 r(t)}{dt^2} + 5 \frac{dr(t)}{dt} + 6r(t) = \frac{de(t)}{dt} - 2e(t)$$

Given $e(t) = u(t)$, $r(0_-) = 1$, $r'(0_-) = 2$, find $r(0_+)$ and $r'(0_+)$.

Solution: $r(0_+) = 1, r'(0_+) = 3$

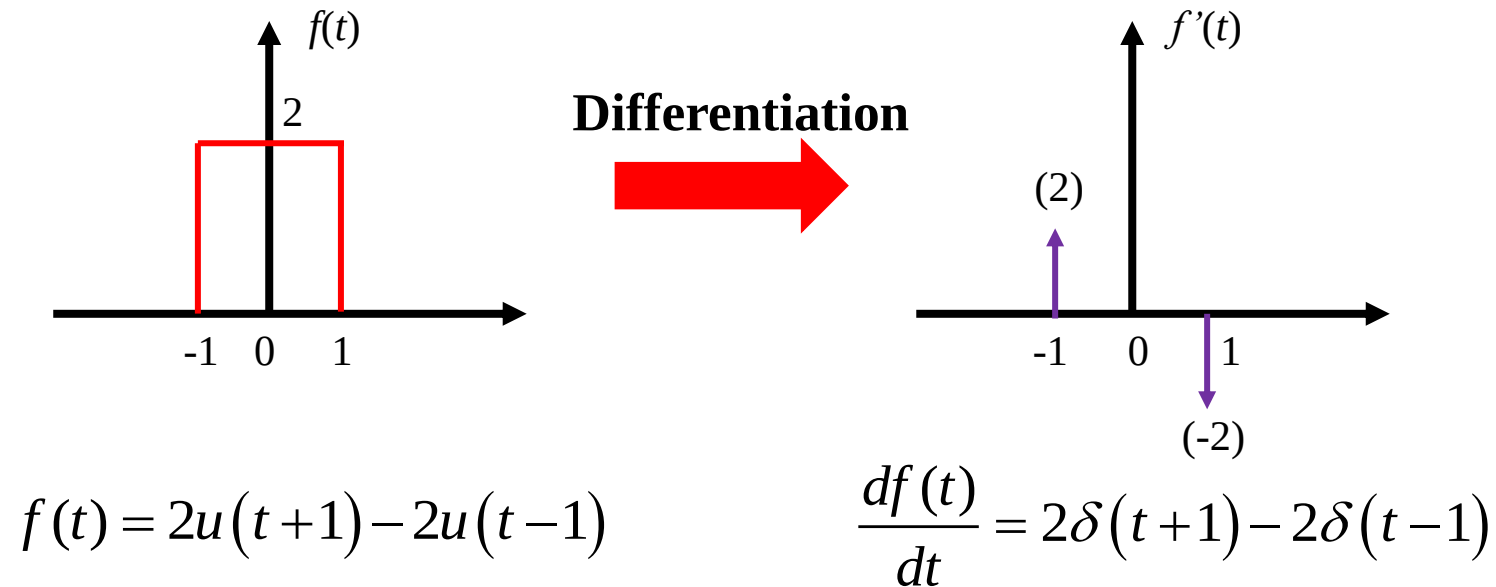
Review of Last Lecture

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

2.5 Zero-Input Response and Zero-State Response

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Impulse Function Matching Method: Impulse functions are the cause of state jumps (for example, an impulse current in a capacitor leads to a jump in its voltage). Therefore, **at $t=0$, the $\delta(t)$ term and its derivatives of all orders on both sides of the differential equation must be balanced.**



If there is a state jump, its derivative must be an impulse function.

In general, the coefficients of the homogeneous solution can be solved by the impulse function matching method.

2.5 Zero-Input Response and Zero-State Response

1. Definition of Zero-Input Response and Determination of Undetermined Coefficients

① Definition: The response generated entirely by the initial state without the action of an external excitation signal, i.e., $r_{zi}(t) = H[\{x(0_-)\}]$

② Equation Satisfied:
$$c_0 \frac{d^n}{dt^n} r_{zi}(t) + \dots + c_{n-1} \frac{d}{dt} r_{zi}(t) + c_n r_{zi}(t) = 0$$

Therefore, it is in the form of a homogeneous solution, i.e.,

$$r_{zi}(t) = \sum_{k=1}^n A_{zik} e^{\alpha_k t}$$

where $\alpha_1, \alpha_2, \dots, \alpha_n$ are n distinct system characteristic roots.

③ Initial Conditions: $r_{zi}^{(k)}(0_+) = r_{zi}^{(k)}(0_-) = r^{(k)}(0_-)$

That is, the undetermined coefficients of the homogeneous solution $r_{zi}(t)$ can be determined by $r^{(k)}(0_-)$.

2.5 Zero-Input Response and Zero-State Response

2. Definition of Zero-State Response and Determination of Undetermined Coefficients

- ① Definition: The response generated only by the excitation when the initial state is zero.

$$r_{zs}(t) = H[e(t)]$$

- ② Equation Satisfied: $c_0 \frac{d^n}{dt^n} r_{zs}(t) + \dots + c_{n-1} \frac{d}{dt} r_{zs}(t) + c_n r_{zs}(t) = E_0 \frac{d^m}{dt^m} c(t) + \dots + E_m$

Therefore, $r_{zs}(t) = \sum_{k=1}^n A_{zsk} e^{\alpha_k t} + r_p(t)$, i.e., (It contains both the homogeneous solution and the particular solution.)

- ③ Initial Conditions: Since the k -th ($k = 0, \dots, n-1$) order derivatives of $r_{zs}(t)$ at time 0^- are all zero, i.e.,

$$r^{(k)}(0_-) = r_{zi}^{(k)}(0_-) + r_{zs}^{(k)}(0_-) = r_{zi}^{(k)}(0_-)$$

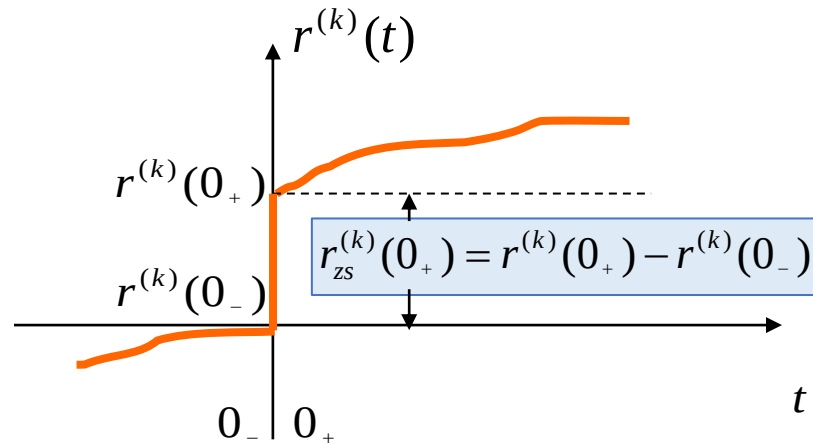
$$r_{zs}^{(k)}(0_-) = 0$$

$$r^{(k)}(0_+) = r_{zi}^{(k)}(0_+) + r_{zs}^{(k)}(0_+) = r_{zi}^{(k)}(0_-) + r_{zs}^{(k)}(0_+)$$

$$r_{zs}^{(k)}(0_+) = r^{(k)}(0_+) - r^{(k)}(0_-) = \text{Jump value}$$

Thus, A_{zsk} is determined by the jump value.

2.5 Zero-Input Response and Zero-State Response



$r^{(k)}(0_+)$: Used to determine the coefficients of the **complete response**

$r^{(k)}(0_-)$: Used to determine the coefficients of the **zero-input response**

$r^{(k)}(0_+) - r^{(k)}(0_-)$: Used to determine the coefficients of the **zero-state response**



Important!

$$\begin{aligned}
 r(t) &= \underbrace{\sum_{k=1}^n A_{Zik} e^{a_k t}}_{\text{zero-input response}} + \underbrace{\sum_{k=1}^n A_{Zsk} e^{a_k t}}_{\text{zero-state response}} + r_p(t) \\
 &= \underbrace{\sum_{k=1}^n (A_{Zik} + A_{Zsk}) e^{a_k t}}_{\text{free response}} + \underbrace{r_p(t)}_{\text{forced response}}
 \end{aligned}$$

2.5 Zero-Input Response and Zero-State Response

Key Facts

$$\text{Complete Response} = \text{Zero-Input Response} + \text{Zero-State Response}$$

Decomposing the response into zero-input response and zero-state response helps to understand the **superposition** and **homogeneity** characteristics of linear systems:

- The **zero-input response** is **linear with respect to the initial state** (initial energy storage) of the system;
- The **zero-state response** satisfies **linearity and time-invariance with respect to the excitation (input signal)** of the system.

Given a linear time-invariant system, under the same initial conditions, when the excitation is $e(t)$, the complete response is $r_1(t) = [2e^{-3t} + \sin(2t)]u(t)$. Let $r_{zi}(t)$ denote the zero-input response and $r_{zs}(t)$ the zero-state response. When the excitation is $2e(t)$, the complete response is $r_2(t) = [e^{-3t} + 2\sin(2t)]u(t)$. Which of the following is correct? ()

- ☐ A $r_{zi}(t) = 2e^{-3t}u(t), r_{zs}(t) = \sin(2t)u(t)$
- ☐ B $r_{zi}(t) = 4e^{-3t}u(t), r_{zs}(t) = [-2e^{-3t} + \sin(2t)]u(t)$
- ☐ C $r_{zi}(t) = e^{-3t}u(t), r_{zs}(t) = 2\sin(2t)u(t)$
- ☒ D $r_{zi}(t) = 3e^{-3t}u(t), r_{zs}(t) = [-e^{-3t} + \sin(2t)]u(t)$

2.5 Zero-Input Response and Zero-State Response

Example 2-7: Given a linear time-invariant system, under the same initial conditions, when the excitation is $e(t)$, the complete response is $r_1(t) = [2e^{-3t} + \sin(2t)]u(t)$; when the excitation is $2e(t)$, the complete response is $r_2(t) = [e^{-3t} + 2\sin(2t)]u(t)$.

- (1) Find the zero-input response and $r_{zi}(t)$ the zero-state response $r_{zs}(t)$.
- (2) If the initial conditions double and the excitation is $e(t-t_0)$ ($t_0 > 0$), find the complete response $r_3(t)$.

Solution: (1) We have

$$\begin{cases} r_1(t) = r_{zi}(t) + r_{zs}(t) = [2e^{-3t} + \sin(2t)]u(t) \\ r_2(t) = r_{zi}(t) + 2r_{zs}(t) = [e^{-3t} + 2\sin(2t)]u(t) \end{cases} \Rightarrow \begin{cases} r_{zi}(t) = 3e^{-3t}u(t) \\ r_{zs}(t) = [-e^{-3t} + \sin(2t)]u(t) \end{cases}$$

- (2) If the initial conditions double and the excitation is $e(t-t_0)$, the complete response is

$$\begin{aligned} r_3(t) &= 2r_{zi}(t) + r_{zs}(t-t_0) \\ &= 6e^{-3t}u(t) + [-e^{-3(t-t_0)} + \sin(2t-2t_0)]u(t-t_0) \end{aligned}$$

Contents of This Lecture

2.6 Impulse Response and Step Response

2.7 Convolution

2.8 Properties of Convolution

Learning Objectives of This Lecture

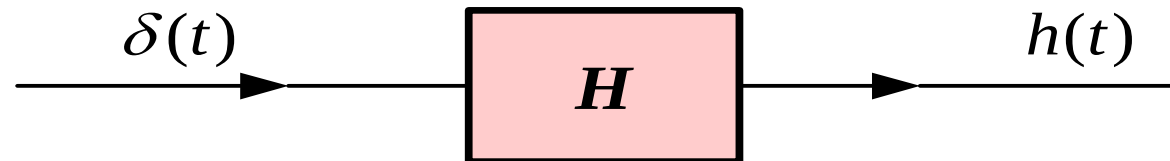
1. Skillfully apply the impulse function matching method to find the **impulse response**;
2. Be familiar with the relationship between impulse response and step response;
3. Understanding the physical meaning of **convolution**;
4. Proficiency in using the **convolutional method** to solve the **zero-state response**;

2.6 Impulse Response and Step Response

2.6.1 Impulse Response

1. Definition

The **zero-state response** generated by a system under the action of a unit impulse signal $\delta(t)$ is called the **unit impulse response** (referred to as **impulse response** for short), which is generally denoted by $h(t)$.

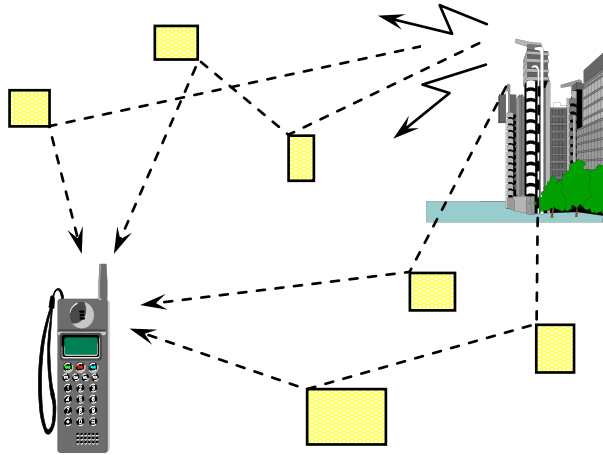


Note: In time domain, for different systems, when the same excitation $\delta(t)$ is applied under zero-state conditions, the resulting responses $h(t)$ are different. This difference indicates distinct system characteristics. Thus, **impulse response can be used to measure a system's properties.**

2.6 Impulse Response and Step Response



Think: How to characterize a wireless multipath channel using impulse response?



2.6 Impulse Response and Step Response

2. Mathematical Model of Impulse Response

An LTI system can be described by a differential equation:

$$C_0 \frac{d^n r(t)}{dt^n} + C_1 \frac{d^{n-1} r(t)}{dt^{n-1}} + \dots + C_{n-1} \frac{dr(t)}{dt} + C_n r(t) = E_0 \frac{d^m e(t)}{dt^m} + E_1 \frac{d^{m-1} e(t)}{dt^{m-1}} + \dots + E_{m-1} \frac{de(t)}{dt} + E_m e(t)$$

Response $r(t)$ and its derivatives (up to the n -th order).

Let $e(t) = \delta(t)$,
then $r(t) = h(t)$

Excitation $e(t)$ and its derivatives (up to the m -th order).

$$C_0 h^{(n)}(t) + C_1 h^{(n-1)}(t) + \dots + C_{n-1} h^{(1)}(t) + C_n h(t) = E_0 \delta^{(m)}(t) + E_1 \delta^{(m-1)}(t) + \dots + E_{m-1} \delta^{(1)}(t) + E_m \delta(t)$$

Which of the following statements are correct? ()

A

Impulse response is the **complete response** when the excitation is a unit impulse function.

B

Impulse response is the **zero-state response** when the excitation is a unit impulse function.

C

Impulse response is the **zero-input response** when the excitation is a unit impulse function.

D

An impulse input signal can be converted into the initial conditions of the system.

Submit

2.6 Impulse Response and Step Response

3. Form of the Impulse Response

$$\begin{aligned} &C_0 h^{(n)}(t) + C_1 h^{(n-1)}(t) + \cdots + C_{n-1} h^{(1)}(t) + C_n h(t) \\ &= E_0 \delta^{(m)}(t) + E_1 \delta^{(m-1)}(t) + \cdots + E_{m-1} \delta^{(1)}(t) + E_m \delta(t) \end{aligned}$$

① Related to the relative orders of n and m

If $n > m$, $h(t)$ does not contain $\delta(t)$ or its derivatives. (**most common**)

If $n = m$, $h(t)$ contains $\delta(t)$ but not its derivatives.

If $n < m$, $h(t)$ contains $\delta(t)$ and its derivatives (up to the $(m-n)$ -th order) .

2.6 Impulse Response and Step Response

3. Form of the Impulse Response

② Related to characteristic roots

If $n > m$ and the characteristic roots are simple (no repeated roots),

$$h(t) = \left[\sum_{i=1}^n A_i e^{\alpha_i t} \right] u(t)$$

For $t > 0_+$, $\delta(t)$ and its derivatives vanish (the excitation no longer exists), so the free term on the right side of the differential equation is zero. At this point:

- The impulse response is a zero-state response with the same form as the zero-input response (equivalent to solving only the homogeneous equation).
- Energy storage introduced by the impulse signal is converted into (equivalent to) the initial conditions of the zero-input response.

2.6 Impulse Response and Step Response

Example 2-8: Given the system differential equation below, find the impulse response $h(t)$.

$$\frac{d^2 r(t)}{dt^2} + 4 \frac{dr(t)}{dt} + 3r(t) = \frac{de(t)}{dt} + 2e(t)$$

Solution: Let $e(t) = \delta(t)$, $r(t) = h(t)$, we have

$$\frac{d^2 h(t)}{dt^2} + 4 \frac{dh(t)}{dt} + 3h(t) = \delta'(t) + 2\delta(t)$$

Step 1: Find characteristic roots

Characteristic equation: $\alpha^2 + 4\alpha + 3 = 0$

Characteristic roots: $\alpha_1 = -1, \alpha_2 = -3$

Step 2: Homogeneous solution form

$$h(t) = (A_1 e^{-t} + A_2 e^{-3t})u(t)$$

2.6 Impulse Response and Step Response

Step 3: Impulse Function Matching Method

$$\frac{d^2 h(t)}{dt^2} + 4 \frac{dh(t)}{dt} + 3h(t) = \delta'(t) + 2\delta(t)$$

Let $\frac{d^2 h(t)}{dt^2} = \delta'(t) + a\delta(t)$, then $\frac{dh(t)}{dt} = \delta(t) + \textcolor{red}{a}\Delta u(t)$, $h(t) = \textcolor{red}{1} \cdot \Delta u(t)$

Substitute them into the differential equation and balance impulses: $a + 4 = 2 \rightarrow a = -2$

Step 4: Determine initial conditions

$$h(0_+) = h(0_-) + 1 = 1, \quad h'(0_+) = h'(0_-) + a = -2$$

Step 5: Solve for the coefficients and the impulse response

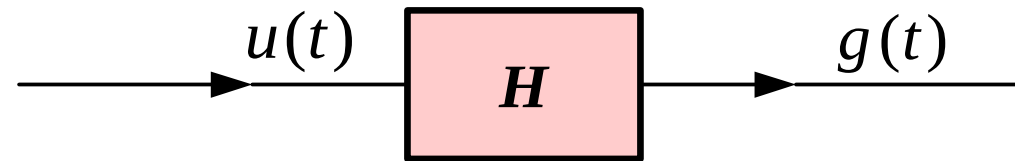
$$\begin{array}{l} h(t) = A_1 e^{-t} + A_2 e^{-3t} \\ h'(t) = -A_1 e^{-t} - 3A_2 e^{-3t} \end{array} \quad \xrightarrow{\text{Set } t=0} \quad \begin{cases} A_1 + A_2 = 1 \\ A_1 + 3A_2 = 2 \end{cases} \quad \longrightarrow \quad \begin{cases} A_1 = 1/2 \\ A_2 = 1/2 \end{cases}$$

$$\text{so } h(t) = \frac{1}{2}(e^{-t} + e^{-3t})u(t).$$

2.6 Impulse Response and Step Response

2.6.2 Step Response

The **zero-state response** generated by a system with the input of a unit step signal $u(t)$ is called the **unit step response** (or step response for short), denoted by $g(t)$.



- The right side of the system equation contains $u(t)$, so the step response includes both a homogeneous solution and a particular solution.
- For LTI systems, the step response can also be found using the relationship between impulse response and step response:

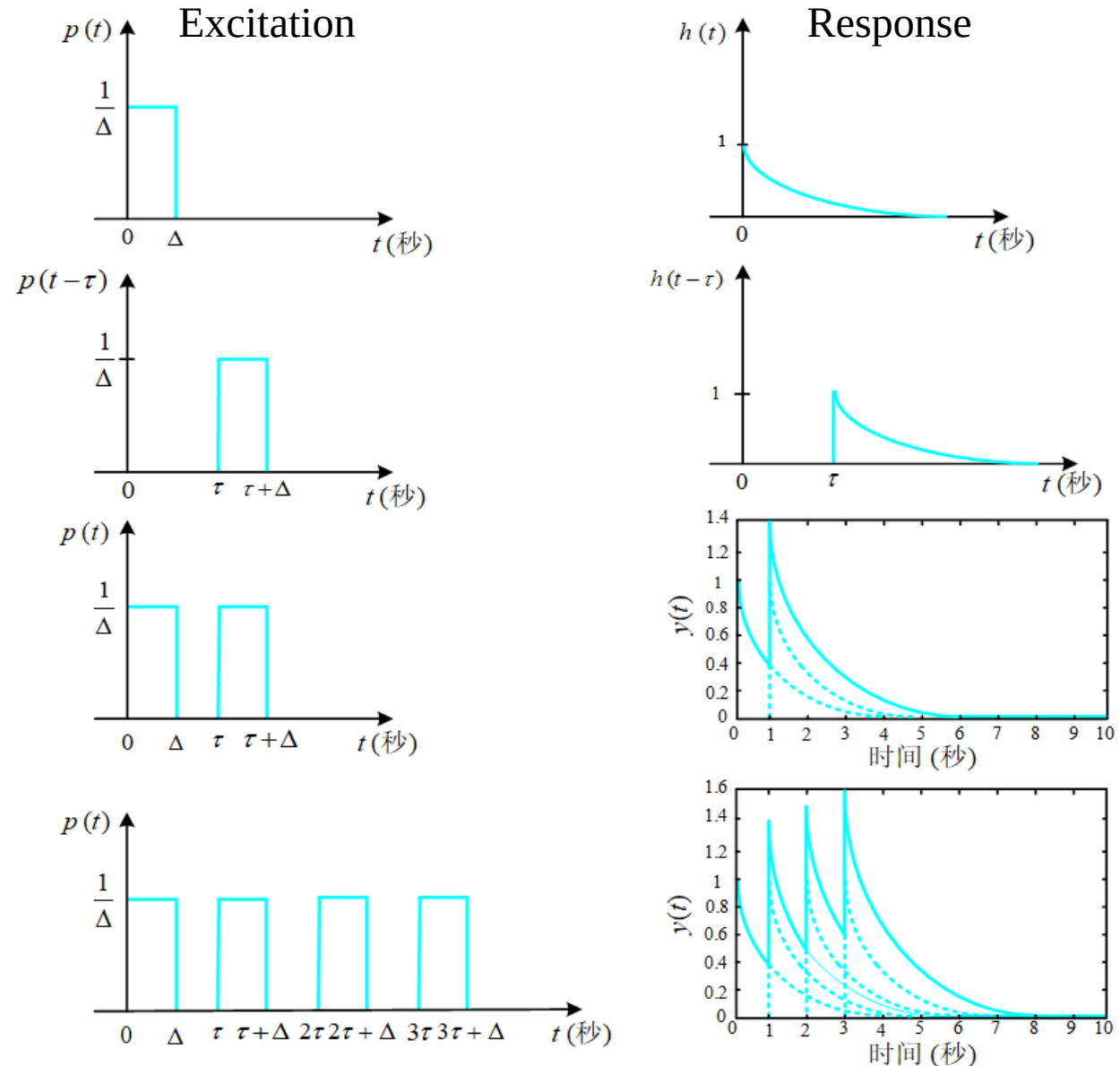
$$g(t) = \int_{0_-}^t h(\tau) d\tau \quad (\text{as } u(t) = \int_{-\infty}^t \delta(\tau) d\tau = \int_{0_-}^t \delta(\tau) d\tau)$$

- The impulse response is the derivative of the step response:

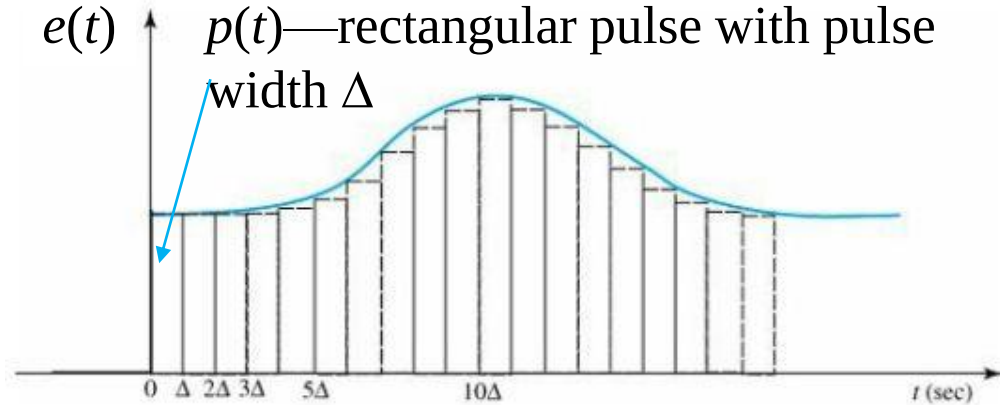
$$h(t) = \frac{dg(t)}{dt}$$

2.7 Convolution

- **Convolution** is a fundamental mathematical tool and a core technique for system analysis.
- The **Convolution Theorem** connects the time-domain analysis and transform-domain analysis of systems, running through the entire course—this makes it extremely important.



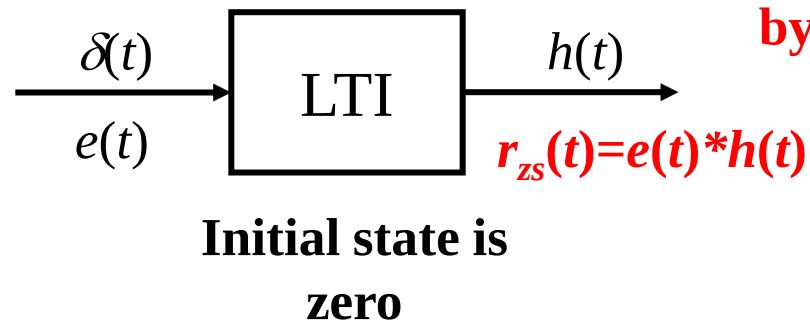
2.7 Convolution



$$e(t) = \lim_{\Delta \rightarrow 0} \sum_{i=0}^{\infty} e(i\Delta) p(t - i\Delta)$$

$$= \lim_{\Delta \rightarrow 0} \sum_{i=0}^{\infty} e(i\Delta) \frac{u(t - i\Delta) - u[t - (i+1)\Delta]}{\Delta} \Delta$$

$$e(t) = \lim_{\Delta \rightarrow 0} \sum_{i=0}^{\infty} e(i\Delta) \delta(t - i\Delta) \Delta = \int_0^{\infty} e(\tau) \delta(t - \tau) d\tau$$



by superposition, homogeneity
and time-invariance

$$r_{zs}(t) = \lim_{\Delta \rightarrow 0} \sum_{i=0}^{\infty} e(i\Delta) h(t - i\Delta) \Delta = \int_0^{\infty} e(\tau) h(t - \tau) d\tau = e(t) * h(t)$$

Convolution

The zero-state response of a system is the convolution of the excitation and the impulse response.

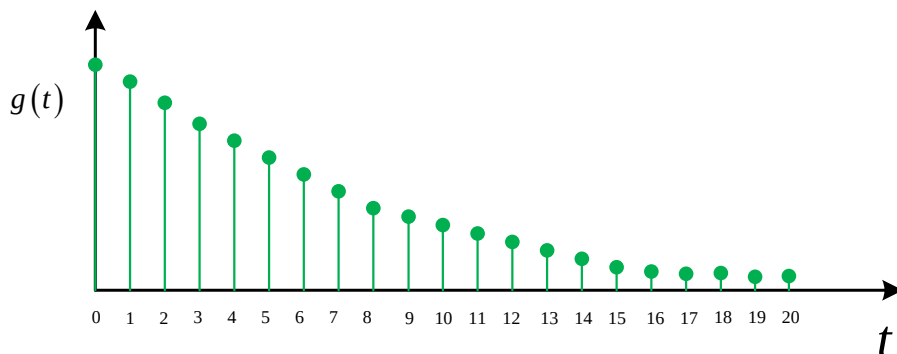
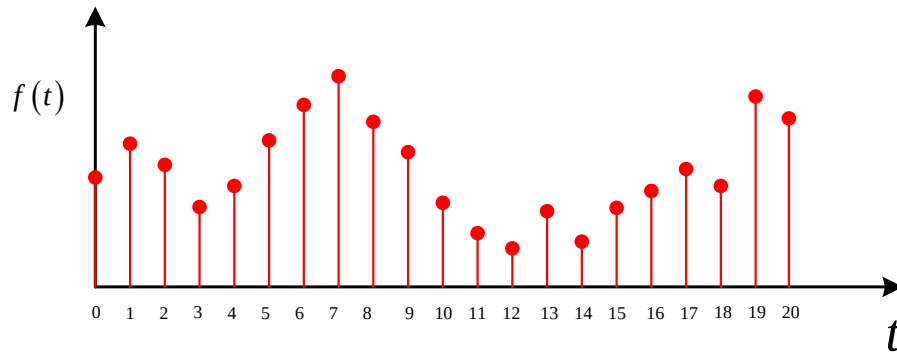
Calculation Method: Decompose any input signal into a weighted sum (integral) of impulse signals, then compute the corresponding weighted sum (integral) of the system's impulse responses.

2.7 Convolution

Definition of Convolution

For two functions $f(t)$ and $g(t)$, the integral $f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau$ is called the **convolution integral** of $f(t)$ and $g(t)$, often abbreviated to "**convolution**" and denoted by the asterisk "*".

Physical Meaning of Convolution

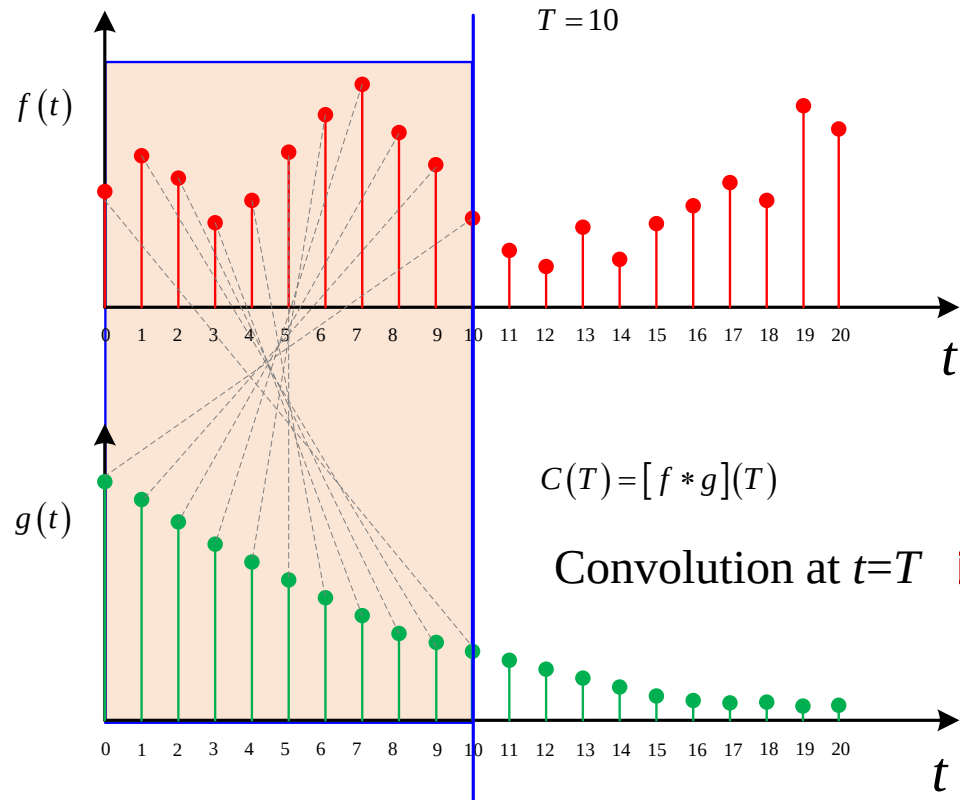


$f(t)$: Input signal of the system.

$g(t)$: Represents the attenuation trend of a signal value (treated as an impulse signal) input at a certain moment.

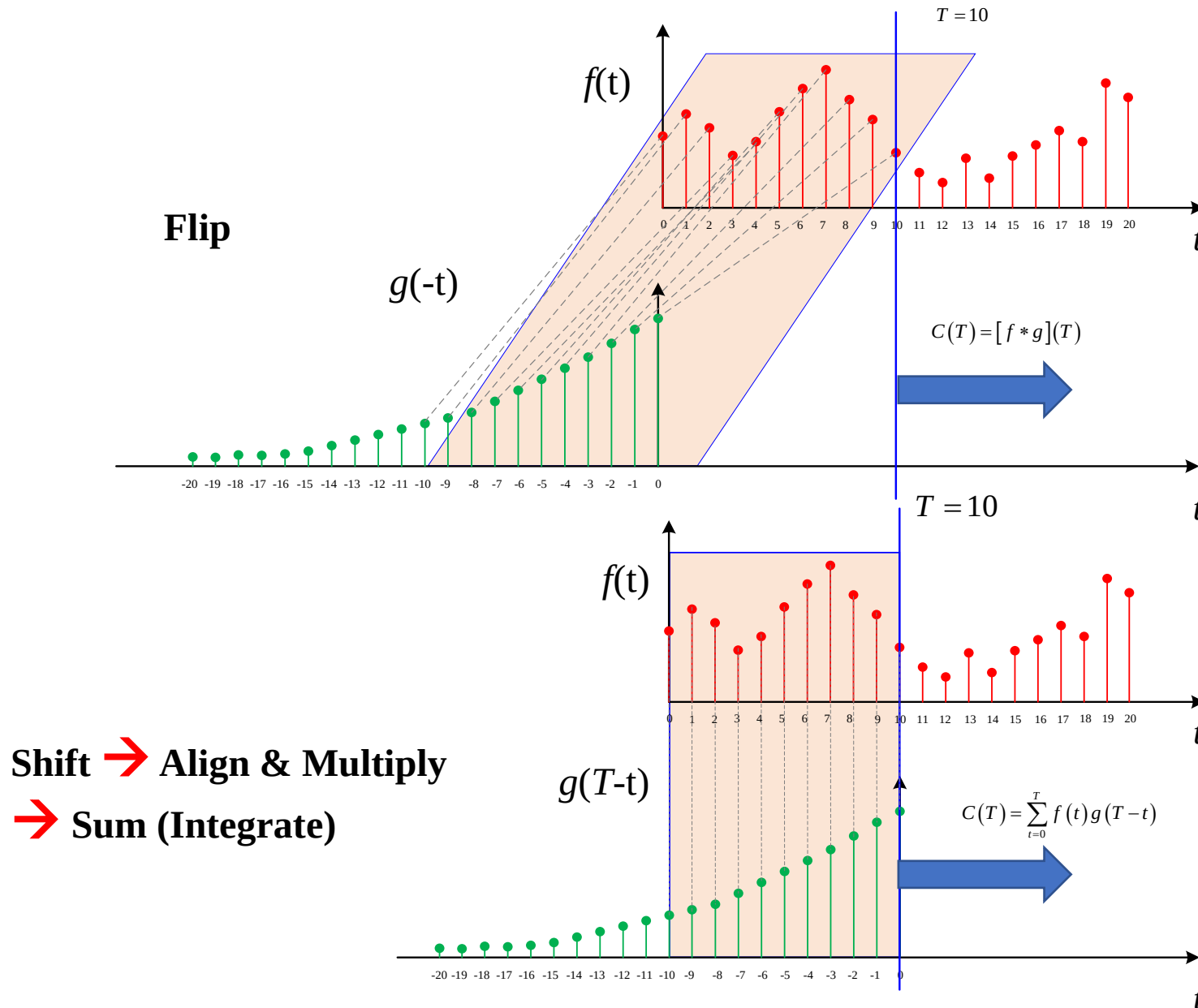
At time $t=T$, the value of the input $f(0)$ (applied at $t = 0$) attenuates to $f(0)g(T)$.

2.7 Convolution



- Since signals are input continuously, the final output of the system reflects the **cumulative effect** of all previous input signals.
- For example, at $t = 10$ (denoted as $T = 10$), the output depends on the entire shaded area in the figure.
 - $f(10)$ is just input at $t = 10$, so its contribution to the output is $f(10)g(0)$;
 - The input $f(9)$ (applied at $t = 9$) has only undergone 1 unit of time attenuation, so its output contribution is $f(9)g(1)$;
 - This pattern continues, as described by the dashed line in the original figure.
- Multiplying these corresponding pairs of values and summing the results gives the output signal value at $T = 10$ —this is exactly the convolution value of $f(t)$ and $g(t)$ at $t = 10$.

2.7 Convolution



Core Steps of Convolution (The "Four-Step Process")

- 1. Flip:** Transform $g(\tau)$ to $g(-\tau)$;
- 2. Shift:** Shift $g(-\tau)$ by t to get $g(t-\tau)$;
- 3. Align and Multiply:** Multiply the overlapping segments of $f(\tau)$ and $g(t-\tau)$;
- 4. Sum (Integrate):** Compute the integral of the product over τ to get the convolution value at time t .

2.7 Convolution

Convolution Operation Analysis for Systems

The steps to calculate the zero-state response for LTI systems by convolution:

1. Replace the independent variable t of both $e(t)$ (input) and $h(t)$ (impulse response) with τ , resulting in $e(\tau)$ and $h(\tau)$.
2. Execute the "four-step process" of convolution:

Flip: $h(\tau) \rightarrow h(-\tau)$

Shift: $h(-\tau) \rightarrow h(t - \tau) = h[-(\tau - t)] \begin{cases} t < 0, & \text{Left shift} \\ t > 0, & \text{Right shift} \end{cases}$

Multiply: $e(\tau)h(t - \tau)$

Integrate: $r_{zs}(t) = e(t) * h(t) = \int_{-\infty}^{\infty} e(\tau)h(t - \tau)d\tau$

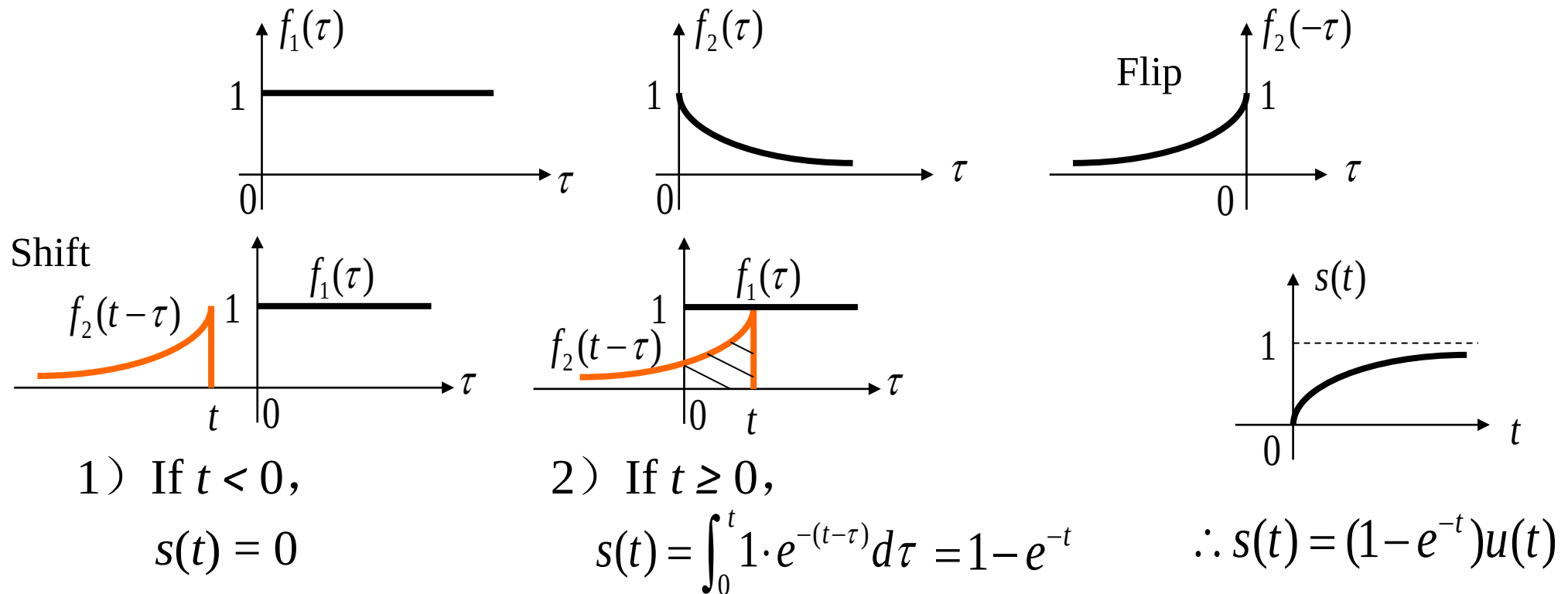
- Key Note:* The most critical part of convolution calculation is determining the **integral limits**—these limits depend on **the intervals where $e(\tau)$ and $h(t - \tau)$ overlap.**

2.7 Convolution

Example 2-9: Given $f_1(t) = u(t)$ and $f_2(t) = e^{-t}u(t)$, find their convolution $s(t) = f_1(t) * f_2(t)$

Method 1: Graphical Method

$$s(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$$



2.7 Convolution

Example 2-9: Given $f_1(t) = u(t)$ and $f_2(t) = e^{-t}u(t)$, find their convolution $s(t) = f_1(t) * f_2(t)$

Method 2: Analytical Method (Direct Integration)

$$s(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau = \int_{-\infty}^{\infty} u(\tau) e^{-(t-\tau)} u(t - \tau) d\tau$$

Step 1: Determine the non-zero intervals of the functions

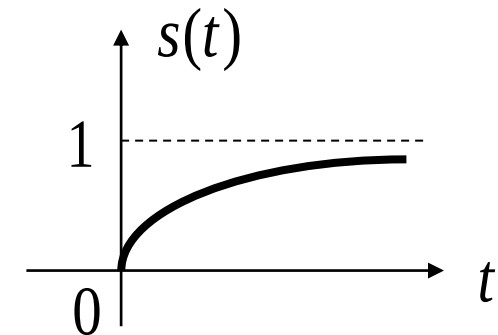
$f_1(\tau) = u(\tau)$ is non-zero only when $\tau > 0$,

$f_2(t - \tau) = e^{-(t-\tau)} u(t - \tau)$ is non-zero only when $\tau < t$.

Step 2: Determine the integral limits by the overlapping

For $t < 0$: No overlap between $\tau > 0$ and $\tau < t$.

For $t \geq 0$: Overlap interval is $0 \leq \tau \leq t$.



$$s(t) = \int_0^t e^{-(t-\tau)} d\tau = e^{-t} \int_0^t e^{\tau} d\tau = e^{-t} (e^t - 1) = 1 - e^{-t} \quad (t \geq 0)$$

Key Note: The analytical method is more efficient for signals with explicit mathematical expressions (e.g., exponential signals, step signals), while the graphical method is better for signals defined by waveforms.

2.7 Convolution

Example 2-10: Given the system's differential equation

$$\frac{d^2 r(t)}{dt^2} + 3 \frac{dr(t)}{dt} + 2r(t) = \frac{de(t)}{dt} + 3e(t)$$

It is known that the excitation $e(t) = e^{-t}u(t)$, and the impulse response is $h(t) = (2e^{-t} - e^{-2t})u(t)$, find the zero-state response.

Solution: The zero-state response of the system is

$$\begin{aligned} r_{zs}(t) &= e(t) * h(t) \\ &= 2 \int_0^t e^{-\tau} e^{-(t-\tau)} d\tau - \int_0^t e^{-\tau} e^{-2(t-\tau)} d\tau \\ &= 2e^{-t} \int_0^t d\tau - e^{-2t} \int_0^t e^{\tau} d\tau \\ &= 2te^{-t} - e^{-2t}(e^t - 1) \\ &= (2t - 1)e^{-t} + e^{-2t} \quad (t > 0) \end{aligned}$$

2.8 Properties of Convolution

Convolution has several fundamental properties that simplify system analysis. For LTI systems, these properties directly reflect the system's inherent characteristics (e.g., time-invariance, linearity).

1. Commutative Law (Exchange Law)

$$f_1(t) * f_2(t) = f_2(t) * f_1(t)$$

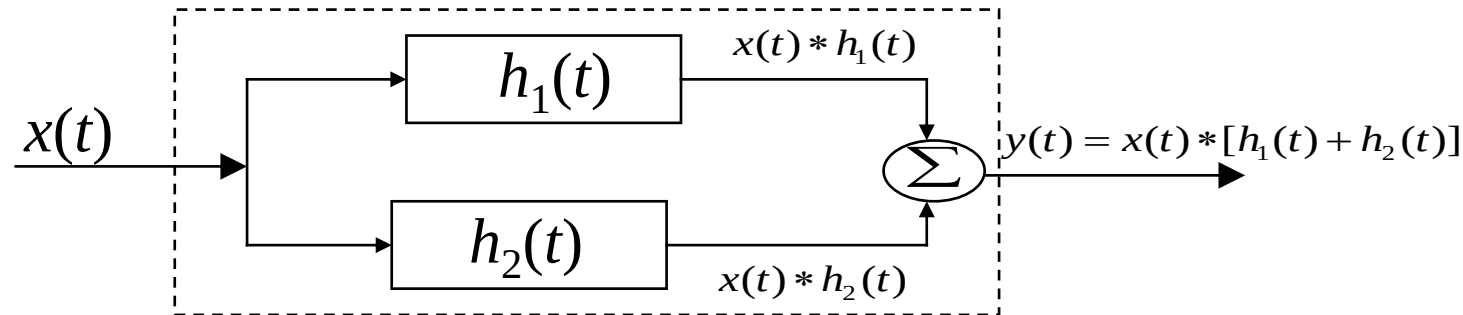
Physical Meaning: The zero-state response of a system with impulse response $h(t)$ to excitation $e(t)$ is identical to the response of a system with impulse response $e(t)$ to excitation $h(t)$.

2.8 Properties of Convolution

2. Distributive Law (Additive Law)

$$f_1(t) * [f_2(t) + f_3(t)] = f_1(t) * f_2(t) + f_1(t) * f_3(t)$$

Physical Meaning: The response of an LTI system to the sum of two excitations equals the sum of the system's responses to each individual excitation. This property directly corresponds to the **superposition principle** of LTI systems.

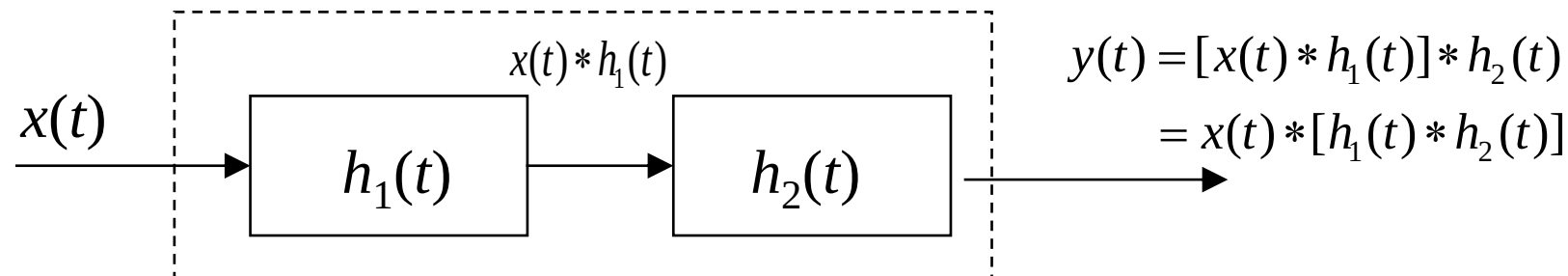


2.8 Properties of Convolution

3. Associative Law

$$[f_1(t) * f_2(t)] * f_3(t) = f_1(t) * [f_2(t) * f_3(t)]$$

Physical Meaning: Cascading two LTI systems (with impulse responses $h_1(t)$ and $h_2(t)$) is equivalent to a single LTI system with impulse response $h(t) = h_1(t) * h_2(t)$. The order of cascading does not affect the final output.



2.8 Properties of Convolution



4. Convolution with Impulse Function

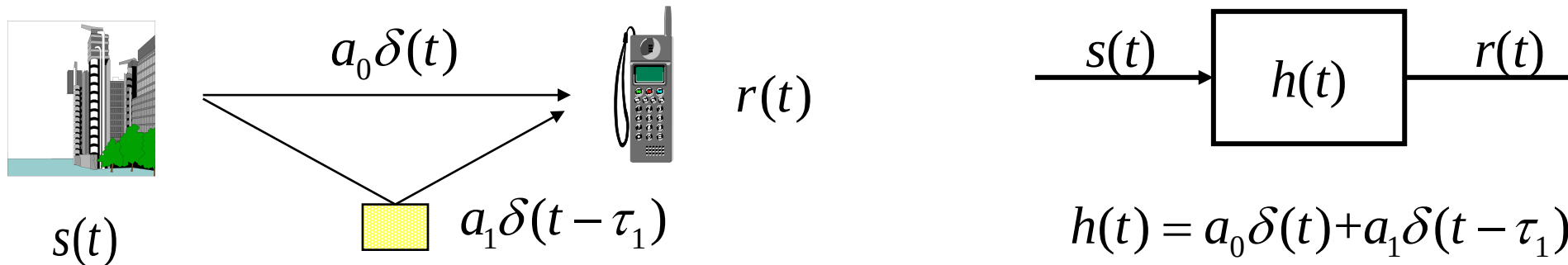
By the sampling property of the impulse function,

$$f(t) * \delta(t) = \int_{-\infty}^{\infty} f(\tau) \delta(t - \tau) d\tau = f(t)$$

$$f(t) * \delta(t - t_0) = \int_{-\infty}^{\infty} f(\tau) \delta(t - \tau - t_0) d\tau = f(t - t_0)$$

Extension: $f(t) * \delta'(t) = f'(t)$ $f(t) * u(t) = \int_{-\infty}^t f(\lambda) d\lambda$

Example: Model the received signal of a wireless communication system with a 2-path channel.



Using the properties of the **associative law** and **convolution with impulse function**, the received signal is

$$r(t) = s(t) * h(t) = s(t) * [a_0 \delta(t) + a_1 \delta(t - \tau_1)] = a_0 s(t) + a_1 s(t - \tau_1)$$

2.8 Properties of Convolution



5. Time-Shift Property

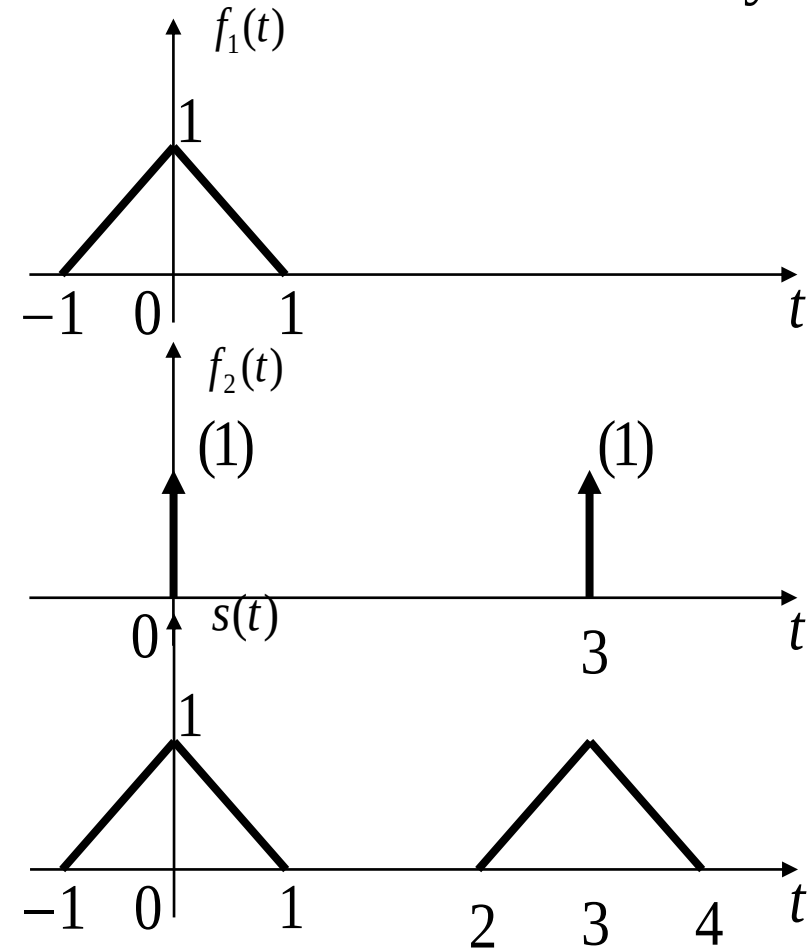
If $f(t) = f_1(t) * f_2(t)$, then $f_1(t-t_1) * f_2(t-t_2) = f_1(t-t_2) * f_2(t-t_1) = f(t-t_1-t_2)$

Physical Meaning: Shifting the excitation by t_1 and the impulse response by t_2 shifts the zero-state response by $t_1 + t_2$ —this reflects the **time-invariance** of LTI systems.

Example 2-11: Given the waveforms of $f_1(t)$ and $f_2(t)$, Find $s(t)=f_1(t)*f_2(t)$ and draw its waveform.

Solution: $f_2(t) = [\delta(t)+\delta(t-3)]$, then

$$\begin{aligned} s(t) &= f_1(t) * [\delta(t) + \delta(t-3)] \\ &= f_1(t) * \delta(t) + f_1(t) * \delta(t-3) \\ &= f_1(t) + f_1(t-3) \end{aligned}$$



6. Time Range of Convolution Integral

Example 2-12: Given that $f_1(t)$ has non-zero values in the interval $t_1 \leq t \leq t_2$ and $f_2(t)$ has non-zero values in $t_3 \leq t \leq t_4$. Let $f(t) = f_1(t) * f_2(t)$. Find the time range where $f(t)$ takes non-zero values.

Solution:

$$f(t) = f_1(t) * f_2(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$$

The integral limits are determined by the **time overlap** between $f_1(\tau)$ and $f_2(t - \tau)$

$$t_1 \leq \tau \leq t_2,$$

$$t_3 \leq t - \tau \leq t_4 \rightarrow t - t_4 \leq \tau \leq t - t_3$$

If $t - t_4 > t_2$ (i.e., $t > t_2 + t_4$), or $t - t_3 < t_1$ (i.e., $t < t_1 + t_3$), there is **no time overlap** between $f_1(\tau)$ and $f_2(t - \tau)$, so the integral equals 0.

Conversely, $f(t)$ has non-zero values in the time range: $t_1 + t_3 \leq t \leq t_2 + t_4$.

The length of this interval is $(t_2 - t_1) + (t_4 - t_3)$, which is the **sum of the lengths of the non-zero intervals** of $f_1(t)$ and $f_2(t)$.

7. Differentiation Property

$$\frac{d}{dt}[f_1(t) * f_2(t)] = f_1(t) * \frac{df_2(t)}{dt} = \frac{df_1(t)}{dt} * f_2(t)$$

Physical Meaning: The derivative of the system's zero-state response equals the response to the derivative of the excitation (or vice versa). This property simplifies the analysis of systems with derivative-based excitations.

8. Integration Property

$$\int_{-\infty}^t [f_1(\lambda) * f_2(\lambda)] d\lambda = f_1(t) * \int_{-\infty}^t f_2(\lambda) d\lambda = \int_{-\infty}^t f_1(\lambda) d\lambda * f_2(t)$$

$$f_1(t) * f_2(t) = \frac{df_1(t)}{dt} * \int_{-\infty}^t f_2(\lambda) d\lambda = \int_{-\infty}^t f_1(\lambda) d\lambda * \frac{df_2(t)}{dt} \quad (f_1(t) \text{ and } f_2(t) \text{ should be 0 when } t \rightarrow -\infty.)$$

Physical Meaning: The integral of the zero-state response equals the response to the integral of the excitation—this complements the differentiation property and is useful for step responses (derived from impulse response integration).

2.8 Properties of Convolution



Commonly used convolutions

$$(1) K * f(t) = K \cdot [\text{area of } f(t)]$$

$$(2) f(t) * \delta(t) = f(t)$$

$$(3) f(t) * \delta'(t) = f'(t)$$

$$(4) f(t) * u(t) = \int_{-\infty}^t f(\tau) d\tau$$

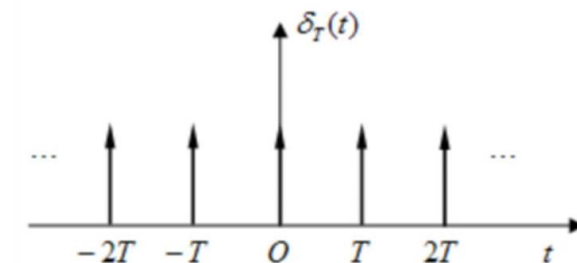
$$(5) u(t) * u(t) = tu(t)$$

$$(6) u(t) * e^{at} u(t) = -\frac{1}{a} [1 - e^{at}] u(t)$$

$$(7) e^{at} u(t) * e^{at} u(t) = te^{at} u(t)$$

$$(8) e^{a_1 t} u(t) * e^{a_2 t} u(t) = \frac{1}{a_1 - a_2} [e^{a_1 t} - e^{a_2 t}] u(t) \quad a_1 \neq a_2$$

$$(9) f(t) * \delta_T(t) = f(t) * \sum_{m=-\infty}^{+\infty} \delta(t - mT) = \sum_{m=-\infty}^{+\infty} f(t - mT)$$



- Repeat the examples in the lecture notes.
- 2.39(a)(b) and 2.48 in the textbook.